

# ME5412 Robotics for Healthcare

## Manipulation aids

操作辅助器具

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## Outline of the lecture

- Target users of manipulation aids
  - Low-tech manipulation aids, for general use
  - Low-tech manipulation aids, for specific use
  - High-tech manipulation aids, for specific use
  - High-tech manipulation aids, for general use
  - Assistive robotic manipulators (ARMs)
- 操作辅助器具的目标用户
  - 通用型低技术操作辅助器具
  - 专用型低技术操作辅助器具
  - 专用型高技术操作辅助器具
  - 通用型高技术操作辅助器具
  - 辅助机械臂 (ARMs)

# 操作辅助器具 目标用户

## Target users of manipulation aids

- 操作：我们通常依靠上肢完成的各类动作，尤其依靠手指与手
  - 大运动操作（幅度更大、发力更强的动作与功能）：够取、抓握/释放、抬举、搬运，以及推、拉、投掷/接住、转动这类动作的协调配合
  - 精细运动操作（幅度更小、更精巧的动作）：捏取、指点，以及手指灵活活动
- **Manipulation**: those activities that we normally accomplish using the upper extremities, particularly the fingers and hand
  - **Gross motor manipulation** (larger and more forceful movement and function): reaching, grasp/release, lifting, carrying, and coordination of movements such as pushing, pulling, throwing/catching, and turning
  - **Fine motor manipulation** (smaller and more delicate movements): pinch, point, and dexterity of finger movements
- The end goal of the integration of these manipulation components is a person's actions in daily and life activities: 将以上各类操作能力结合起来，最终目的是完成人的日常与生活活动：
  - handwriting, food preparation, eating, reaching, and using controls in the environment such as door handles, elevator buttons, and appliance buttons depend on manipulation of physical objects. • 书写、备餐、进食、够取物品，还有操作环境中的各类控件，例如门把手、电梯按钮、家电按键，都离不开对实物的操作。
- **Target users are individuals with upper-limb disabilities:** 目标用户：上肢残障人群
  - Quadriplegic individuals (spinal cord injuries)
  - Individuals with limited motor abilities, in terms of range of motion and/or strength
    - 四肢瘫痪人群（脊髓损伤）
    - 活动范围和/或肌力受限的运动能力障碍人群

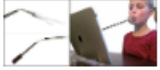
# 操作辅具的分类

## Classification of manipulation aids

|                                         | 低技术<br>Low-tech                                                                      | 高技术<br>High-tech                                                                     |
|-----------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Specific tasks<br>(e.g., feeding)<br>专用 |   |   |
| General use<br>通用                       |  |  |

## Which are low-tech manipulation aids, for general use?

✓ 0



✓ mouth stick

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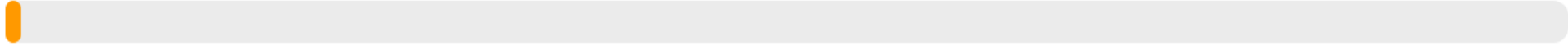
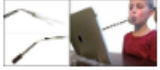


✓ head pointer.

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## Which are low-tech manipulation aids, for specific use?

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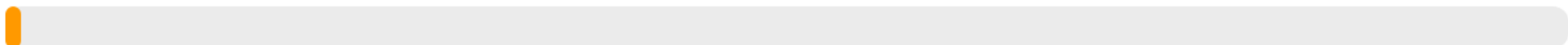
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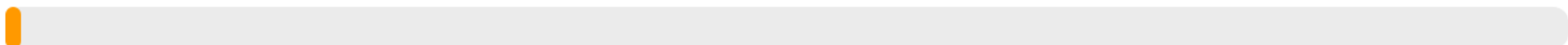
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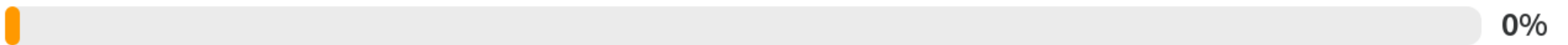
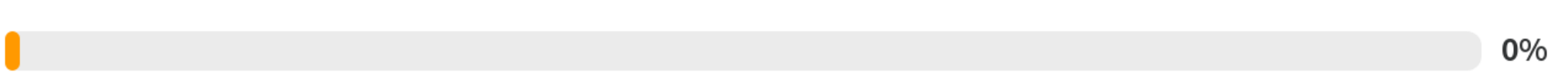
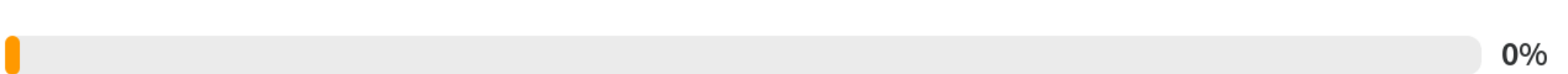
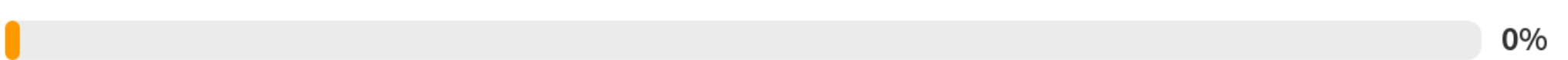
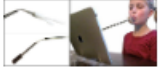
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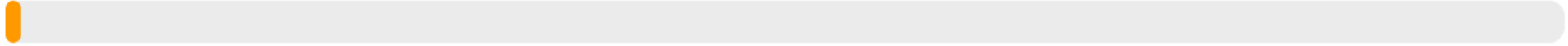
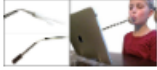
## Which are high-tech manipulation aids, for specific use?

✓ 0

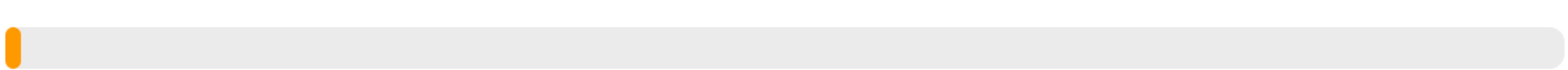


# Which are high-tech manipulation aids, for general use?

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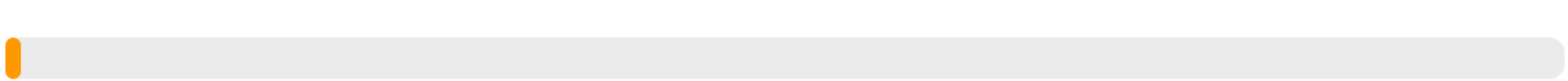


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mobile Robotic

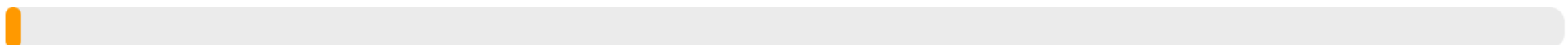


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Wheel-chair mounted Robotic Arm

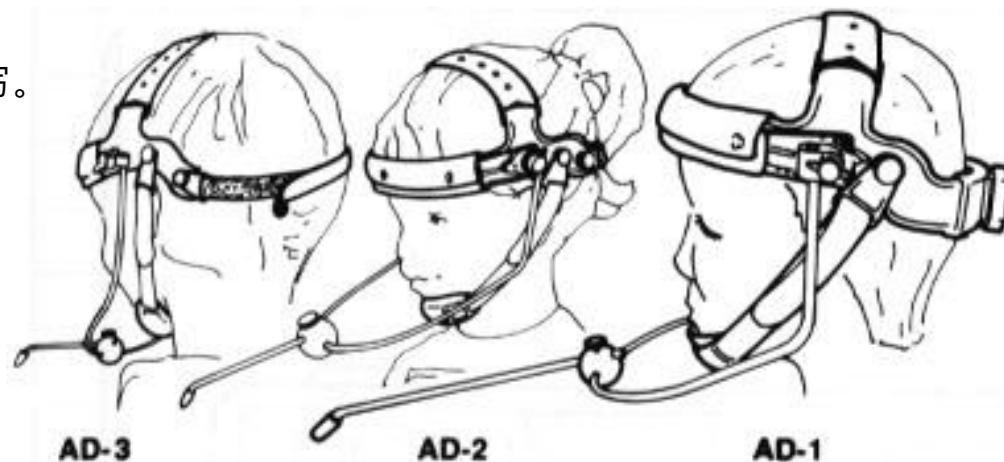


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# 通用型、低技术 操纵辅具

## Low-tech manipulation aids, for general use

- 口控棒：一般是一根木杆，一端可以安装不同配件，另一端是塑形部件，使用者可以用嘴咬住杆件。
- 头控指点棒结构与之类似，区别在于杆件固定在佩戴于前额的头带上。
- A **mouth stick** is usually a wooden dowel with different attachments at one end and a molded piece at the other end that allows the person to hold the stick in mouth.
- A **head pointer** is similar in design, except the dowel is fixed to a headband that is worn around the forehead.
- 两者都依靠头部运动来操控。
- Both are controlled by head movements.
- A pen or a pencil can also be attached to a mouth stick for writing.
- 口控棒末端可以装上钢笔或者铅笔，用于书写。
- Additional attachments include a pincher that is opened or closed by tongue action and a suction cup end that can be used to grip objects (e.g., a page) by sucking on the end of the mouth stick.
- Mouth sticks or head pointers can also be used for such functions as typing and turning lights on and off.
- 其他可选配件：靠舌头控制开合的夹取头；吸盘端头，使用者通过向口控棒吸气，就可以吸附物体（例如书页）。
- 口控棒与头控指点棒还可以完成打字、开关灯等操作。



# Low-tech manipulation aids, for general use

- **Reachers** extend range of motion of

- Wheelchair users
- Individuals that have bending difficulties

取物杆可以拓展以下人群的活动可达范围：

- 轮椅使用者
- 无法弯腰的人群

- A **gripper** on one side and a **handle** with a lever to close the gripper

- Handle grasping may be:

- With full hand
- With all fingers
- With the index finger

- 一端是夹爪，手柄处配有控制夹爪闭合的扳杆
- 握持手柄的方式可以：

- 整只手握住
- 全部手指抓握
- 仅用食指操作

可以用来拿取食品（罐头、包装袋）、厨具（锅、平底锅、盘子、餐具）、办公用品（纸张、书本、杂志），还有休闲物品（书籍、扑克牌、电子阅读器）。

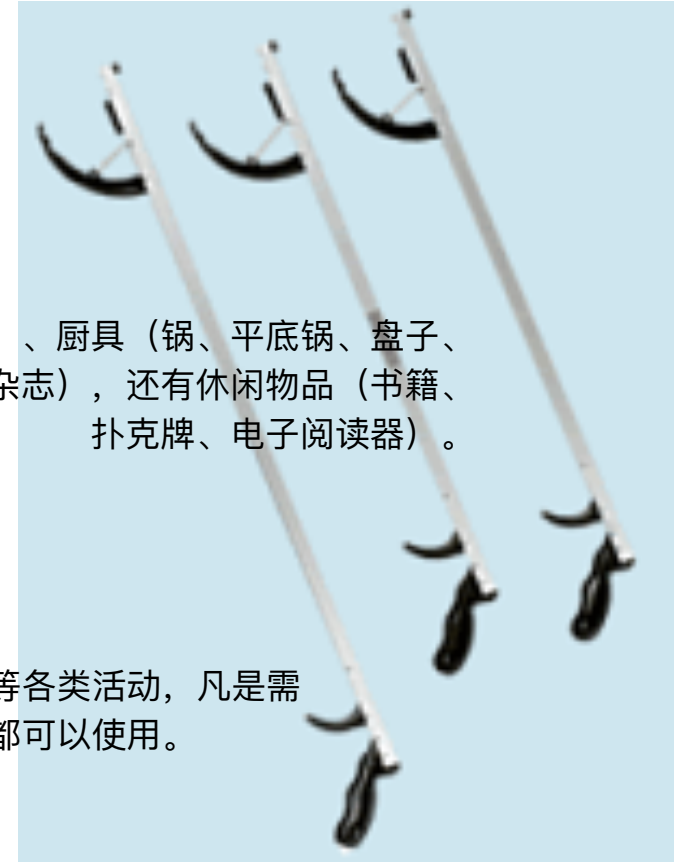
- Used to manipulate food (e.g., cans, packages), cooking utensils (e.g., pans, pots, plates, dishes), office objects (e.g., paper, books, magazines), and recreational or leisure objects (e.g., books, playing cards, eReaders).

- Activities such as dressing, hygiene and other self-care activities, and household activities—any activity that requires grasp and retention of an object that is outside of the individual's reach.

用于穿衣、个人卫生护理以及家务等各类活动，凡是需要抓握、拿取手臂够不到的物体，都可以使用。

- **IMPORTANT:** because a reaching device extends the length of the arm, it also extends the length of the lever arm, resulting in increased force required to lift the object.

⚠重要提示：取物杆会等效加长手臂，也加长了力臂，因此抬举物体时需要更大的力。



专用型，低技术操作辅助

## Low-tech manipulation aids, for specific use

- 仅为一到两项特定任务设计
  - 在对应任务中使用效果很好
  - 但想要满足自理、工作、休闲全部需求，往往需要准备多件器具。

- Designed for one or two tasks only

- they serve those tasks very well
- however, several are needed to meet the demands of self-care, work, and leisure.

- Often standard tools, adapted through: 很多是在标准工具基础上改造而来，改造方式分为四类：

1. lengthening a handle or reducing the reach required 1. 加长手柄，降低对肢体伸展距离的要求
2. modifying the handle of a utensil for easier grasping or manipulation 2. 修改工具手柄，让抓握、操作更加省力
3. converting two-handed tasks to one-handed ones (providing an alternative method of stabilizing the tools that are being used) 3. 将双手操作的任务改为单手可完成（提供替代方案固定工具）
4. amplifying the force that a consumer can generate with hands. 4. 放大使用者手部可以输出的力量

# Low-tech manipulation aids, for specific use

- 产品的专用改造实例：
- 加粗手柄，更容易抓握
  - 固定餐具的束带，可以套住手指
  - 角度弯折手柄，方便手腕活动受限人群舀取食物

Special-purpose adaptations of products include

- Enlarged grips for easier grasping,
- Cuffs that hold a utensil and circle the fingers,
- Angled handles for ease of scooping (for people with limited wrist movement),
- Swivel handles that allow the end to be oriented differently for different positions in space (e.g., on a table or near the mouth),
- Handles requiring limited grasp.

- 旋转手柄，可以根据空间位置调整端头朝向（比如放在桌面，或是送到嘴边）
- 对抓握力度要求更低的手柄



# 专用型, 高技术 操纵

## High-tech manipulation aids, for specific use

翻页器

两项核心动作:

1. 将待翻的页面和其他书页分离开
2. 把这一页从一侧移动到另一侧 (向前翻页 / 向后翻页)

### Page turners

- Two primary actions:

1. separating the page to be turned from the other pages
2. physically moving the page from one side to the other (forward or backward).

- Additional useful but not essential features include

1. scanning several pages
2. turning to a specific page
3. locating a bookmark
4. turning to that page

其他实用但非必需功能:

1. 连续扫描多页
2. 跳转到指定页码
3. 识别书签位置
4. 跳转到书签对应的页面

- Turning a single page is the most difficult task for page turners. Once the page to be turned is successfully isolated, the page turner must move it to the opposite side of the book or magazine. 单页翻页是翻页器最难实现的任务。成功分离目标书页之后, 翻页装置还需要把这一页推送至书本或杂志的对侧。

- Page turners use several different mechanisms: 翻页器常用机构:

- rubber-coated wheel to separate the pages, and then a rotating semicircular disk to push the separated page from one side to the other. As the disk rotates, the page is moved forward or backward, depending on the direction of rotation of the disk.
- a vacuum to suck the top page away from the rest of the book and then turn the page.

- Advances in mainstream page turners have been driven by the need for musicians to have hands-free page turning.

- Typically, page turners use two switch inputs: forward and backward.

- 橡胶涂层滚轮分开书页, 再用旋转半圆拨盘把分离好的书页拨到另一侧; 拨盘旋转方向决定向前还是向后翻页。

- 使用真空吸力把最上方的书页吸起, 再完成翻页。

- 主流翻页设备的技术迭代, 主要来自音乐家对免手翻谱的需求。

- 翻页器一般配备两个开关输入: 向前翻页、向后翻页。



# High-tech manipulation aids, for specific use

## Feeders

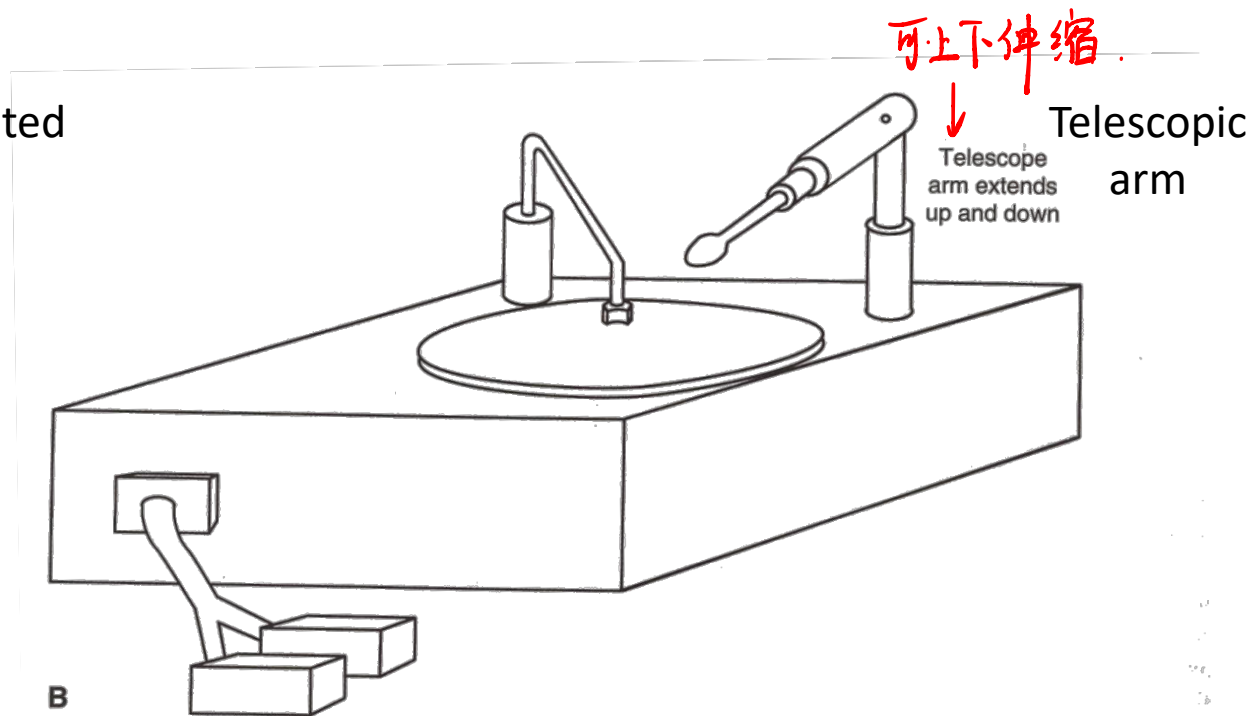
Two functions:

- To locate the food
- To scoop the food

喂食机器人

两大功能：

- 定位食物
- 舀取食物



Food on turntable, the user stops turning when desired food is properly positioned

食物放在转盘上，用户转动转盘，目标食物转到合适位置时停止转动。

# Neater Eater by Neater Solutions Ltd, UK

机械臂具备2个自由度：

- 上下运动
- 前后运动

2 DOF in the arm:

- Up/down
- Forward/backward

1 DOF in the spoon

- Rotation
- 勺子具备1个自由度：

- 旋转



<https://www.neater.co.uk/neater-eater>

<https://youtu.be/Nm5XZG059Ew?si=6SQF7IGCjhjbAcB->

# Neater Eater Control

勺子的4种控制模式：

4 control modes for the spoon:

1. One cycle: the spoon moves once
2. The spoon moves only if the switch is pressed
3. The spoon moves continuously
4. Single cycle: it starts when dish rotation is also activated

The cup is not grasped but the user drinks from the straw

杯子不需要机械夹持，使用者通过吸管饮用。



1. 单次循环：勺子完成一次完整动作
2. 仅按下开关时勺子才运动
3. 勺子持续循环运动
4. 单循环模式：餐盘旋转启动的同时，勺子动作启动

# Handy 1 by RehabRobotics, UK

- Programmable robotic arm
- 5 DOF

- 可编程机械臂
- 5个自由度
- 重量：25千克
- 高度：99厘米-120厘米
- 宽度：76厘米



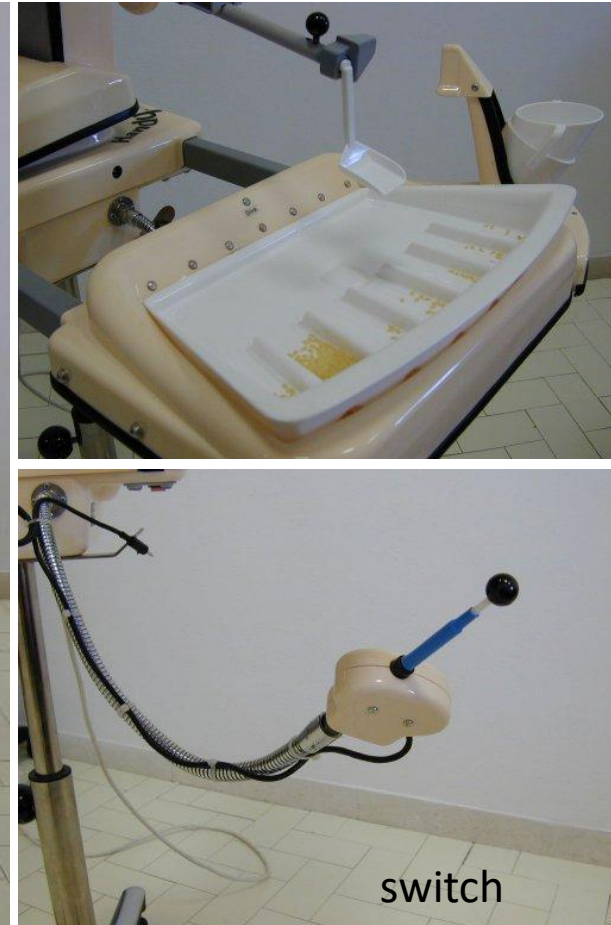
**Weight:** 25 Kg  
**Height:** 99 cm ÷ 120cm  
**Width:** 76 cm

# Handy 1 by RehabRobotics, UK

- Programmable robotic arm
- 5 DOF
- Indirect user interface
- Scanning system (LED)
- 7-slot tray
- Cup holder
- Wheeled cart
- On-board control system



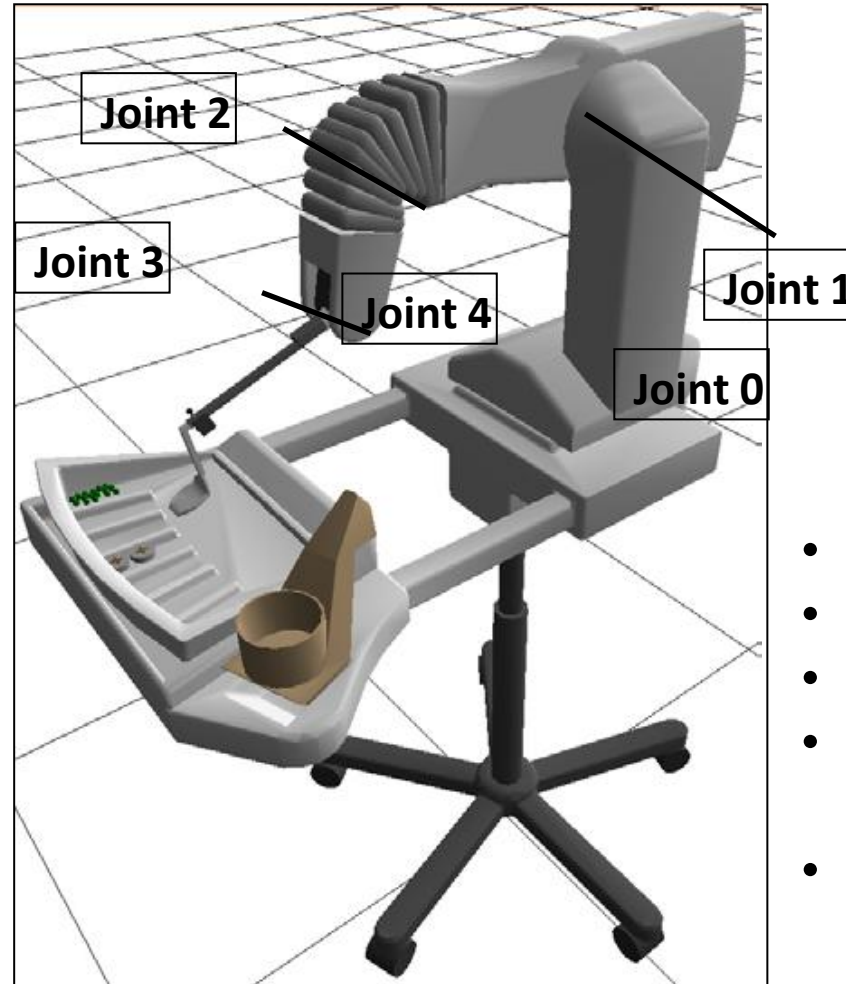
- 间接式用户交互界面
- LED扫描选择系统
- 7格物品托盘
- 杯架
- 可移动底座推车
- 机载控制系统



**Weight:** 25 Kg  
**Height:** 99 cm ÷ 120cm  
**Width:** 76 cm

# Handy 1 by RehabRobotics, UK

- Programmable robotic arm
- 5 DOF
- Indirect user interface
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使用者通过扫描选择托盘的格子  
勺子移动到目标格子  
勺子向前伸出舀取食物  
勺子继续运动，一共有3个目标停靠位置  
机器人内置  $3 \times 7 + 1$ （杯子）组预设位置

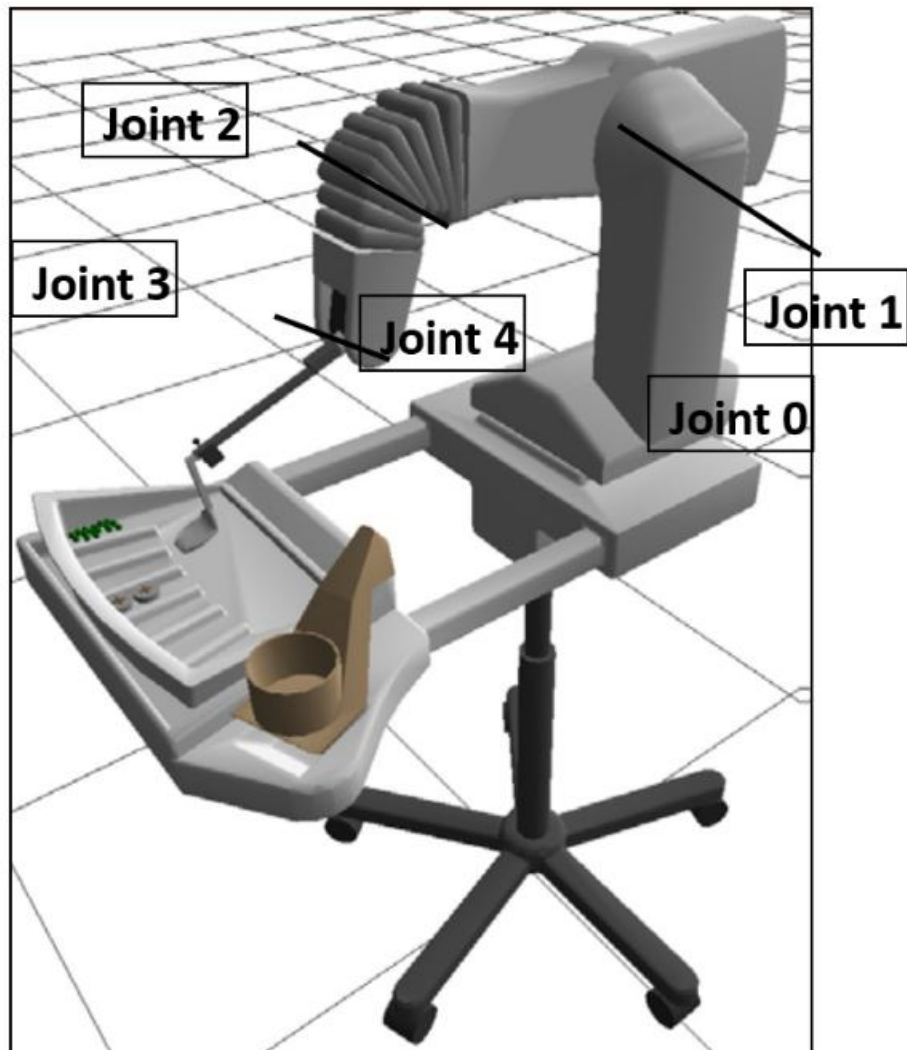


- The user selects the tray slot by scanning
- The spoon reaches the slot
- The spoon moves forward and scoops food
- The spoon moves further, next times (3 positions in total)
- The robot uses  $3 \times 7 + 1$  (cup) pre-set positions

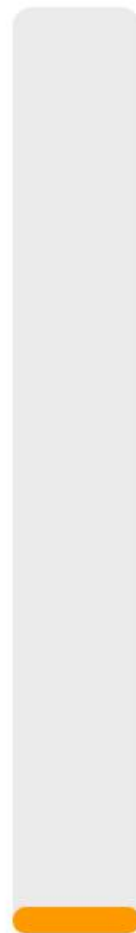
## How to program Handy1: Is it a structured environment?

✓ 0

固定式

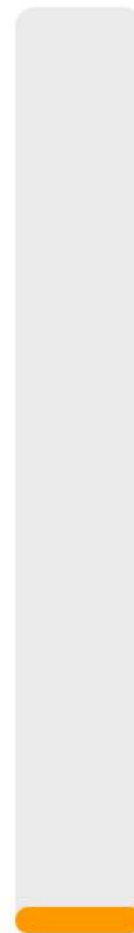


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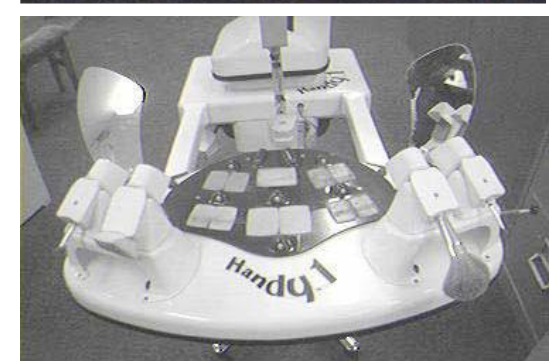


False

# Handy 1 tasks

Handy1可执行任务

- **Feeding Set:** LED-based scanning allows the user to select the food slot
- **Cleansing Set:** shaver, toothbrush and cup
- **Make-up Set:** cosmetic products to apply on face
  - 进食套件：依靠LED扫描，用户选择盛放食物的托盘格
  - 清洁套件：剃须刀、牙刷、水杯
  - 化妆套件：各类面部化妆用品



通用型, 高技术 操辅

# High-tech manipulation aids, for general use

From intro  
lecture

**Robotic Assistive Technologies**

Personal service robots

Assistive Robots (AR)

Socially Assistive Robots (SAR)

Socially Interactive Robots (SIR)

Robotic Assistive Technologies

A sub-class of Assistive Robots (AR), because they are used by individuals with disabilities (while AR are used by any user)

- Robotic prosthetic arms and legs
- Robotic manipulators
- Bionic arms
- Exoskeletons

P. Encarnação, A.M. Cook (2017) *Robotic Assistive Technologies: Principles and Practice*, CRC Press

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# High-tech manipulation aids, for general use

辅助机械臂

## Assistive robotic manipulators – ARMs

### Differences with industrial robotic manipulators:

- Limited speed and force
- Not in a fixed position
- Human in the control loop

和工业机械臂的区别：

- 速度、输出力受限
- 并非固定安装
- 人处于控制回路之中

### Differences with prostheses:

- Not attached to the human body

和假肢的区别：

- 不佩戴、不附着在人体身上

# Assistive robotic manipulators – ARMs

Definition: 辅助机械臂 (ARMs) 定义

“Any robotic arm that is not body bound with a minimum of 6 DOF and a gripper, which is safe to interact with humans, especially humans with special needs or a disability, and can assist with several activities under the control of the user”

一种非人体附着的机械臂，至少具备6个自由度与夹持器；可以安全与人交互，尤其是面向特殊需求、残障人群；在使用者控制下可以完成多项活动。

# Example of drinking task

Required DOF for Drinking

完成喝水动作所需要的自由度

|   | Action                | Robot Joint Actions                                                                                                                                    | Required DOF        |
|---|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1 | Robot holding a glass | Starting point                                                                                                                                         |                     |
| 2 | Robot lifts glass     | z-axis movement up and simultaneously movement of gripper down (pitch) to keep glass horizontal                                                        | $z + \text{pitch}$  |
| 3 | Move glass to user    | x-axis and y-axis movement toward the user and simultaneous rotation of the gripper on the horizontal plane (yaw) to keep the same gripper orientation | $x, y + \text{yaw}$ |
| 4 | Start drinking        | z-axis movement and roll of gripper                                                                                                                    | $z + \text{roll}$   |

| 动作                                               | 机器人关节动作   | 需要自由度 |
|--------------------------------------------------|-----------|-------|
| 1 机器人握住杯子，处于初始位置                                 | -         | -     |
| 2 机器人抬起杯子：沿Z轴向上，夹持器同时俯仰，保证杯子保持水平                 | Z轴 + 俯仰   | 2     |
| 3 将杯子移动到使用者面前：X轴、Y轴朝向使用者运动；夹持器水平面偏转（偏航），维持夹持姿态不变 | X、Y轴 + 偏航 | 3     |
| 4 开始饮水：Z轴运动 + 夹持器滚转倾斜杯子                          | Z轴 + 滚转   | 2     |

# Target users of ARMs

手臂、手部功能完全丧失或部分丧失的人群。

依据国际功能、残疾和健康分类（ICF）：

Anyone who has no or limited arm or hand function

According to the ICF:

- People with amyotrophic lateral sclerosis (ALS)
  - People with muscular dystrophy (MD)
  - People with spinal muscular atrophy (SMA)
  - People with spinal cord injuries (SCIs)
  - Stroke survivors
  - People with strong tremor/spasticity
- 肌萎缩侧索硬化（ALS）患者
  - 肌营养不良（MD）患者
  - 脊髓性肌萎缩症（SMA）患者
  - 脊髓损伤（SCI）患者
  - 中风幸存者
  - 存在严重震颤、肌肉痉挛的人群

# Target users of ARMs

- 使用需求：当使用者明确感受到设备带来的实际价值时，接受度会提高。老年人选择产品更看重功能；年轻人会更多考虑外观。
- 性别：男性更看重设备技术参数；女性更看重机器人交互带来的情感体验。

**Elderly people** constitute another group of potential users of robots that assist in manipulation

Functional decline may create the need for a robot able to retrieve objects for its user

**Individual factors** affecting **acceptance** of healthcare robots by older people include the following:

- **Age:** Older people are less likely to imagine living daily with robots. However, acceptance of robots as a way of gaining independence increases in the oldest age range;
- **Needs:** Acceptance of robots by older people increases when there is a perceived need for the device. This is in line with the fact that older people tend to use products because of their functional aspects, in contrast to younger people who often use products because of stylistic considerations regardless of functionality
- **Gender:** Males value more technical characteristics, and females value the emotional aspects of robot interaction
- The more **experience** users have with technology and robots, the greater their level of acceptance
  - 技术使用经验：接触机器人、科技产品越多，接受度越高。
- There is some evidence suggesting that acceptance of robots increases with **cognitive ability** and **level of education**.
  - 认知水平与受教育程度越高，对机器人的接受度通常越高。

老年人也是操作辅助机器人的潜在用户群体。  
身体机能衰退，会产生机器人代为取物的需求。

影响老年人对医疗机器人接受度的个体因素：

- 年龄：年纪偏大的人较难想象日常使用机器人；但是高龄群体，为了获得独立生活能力，接受度会上升。

# 目标用户的条件

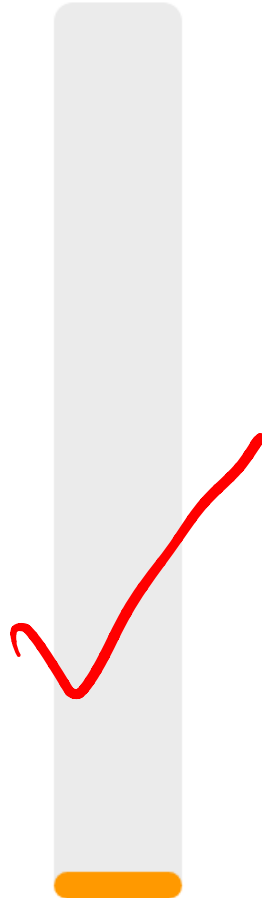
## Target users of ARMs

- Strong need for independence
  - Good cognitive skills for path planning
  - Good cognitive skills for learning the control menu
  - Abilities for controlling ARM movements
  - Vision
  - Powered-wheelchair users, so as to provide battery power supply to the ARM
- 强烈的独立生活意愿
  - 具备不错认知能力，可以规划动作路径
  - 具备不错认知能力，可以学习控制菜单
  - 拥有可以操控机械臂的能力
  - 具备视觉
  - 使用电动轮椅（可以为机械臂提供电池供电）

Tom is 40-year old, had a car accident recently, leaving him quadriplegic. He leaves alone, he is well-educated and has a rewarding professional career. Is Tom a potential ARM user?



0%



True

0%

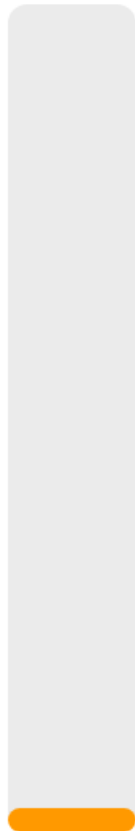


False

Rose is 83 year old and recently suffered from stroke, leaving her emiplegic. She started having mild cognitive issues. She leaves with her husband and has little familiarity with technology. Is Rose a potential ARM user?



0%



True

0%



False

# ARM tasks

- Grasping and handling objects
- ADLs like:
  - Bathing and showering
  - Dressing
  - Self-feeding and –drinking
  - Personal hygiene and care (including combing hair, brushing teeth)
  - Toilet hygiene (getting to the toilet, cleaning oneself, getting back up)
  - Doing housework
  - Preparing meals
  - Shopping
  - Using telephone or other form of communication

- 抓取、搬运物体
- 日常生活活动:
  - 洗浴
  - 穿衣
  - 自主进食、喝水
  - 个人清洁护理（梳头、刷牙）
  - 如厕护理（去到卫生间、清洁、起身）
- 家务劳动
- 准备餐食
- 购物
- 使用电话和其他通讯设备

辅助机械臂控制方式

# ARM control

- Let's consider a joystick as an interface → 2 control signals
  - see HTIs for adaptation and access to the joystick
- Control of  $n$  DOFs, plus a gripper

以操纵杆作为交互设备举例：仅输出两路控制信号。

（可以借助人机接口硬件适配操纵杆，方便残障用户访问）

要控制 $n$ 个自由度，外加夹持器。

## How to control the ARM?

如何控制

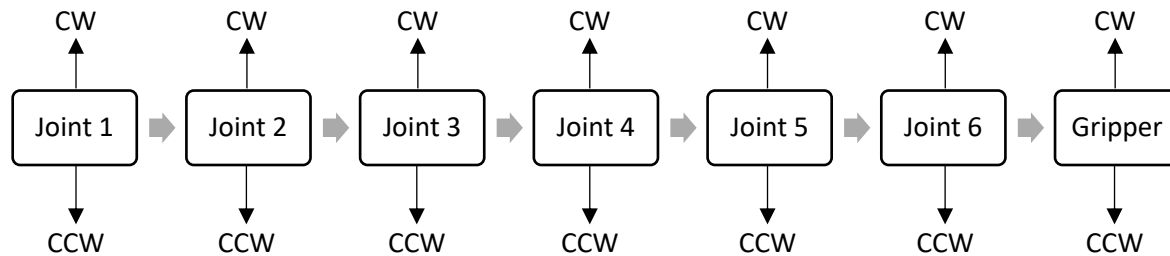
# ARM control: joint vs. Cartesian

## Joint control

关节空间控制

- 一次控制一个自由度
- 需要训练，理解关节转动和笛卡尔空间目标位置的映射关系
- 依靠正向运动学

- DOFs are controlled one at a time
- Some training is needed for mapping joint rotations to Cartesian desired position
  - **Forward kinematics**

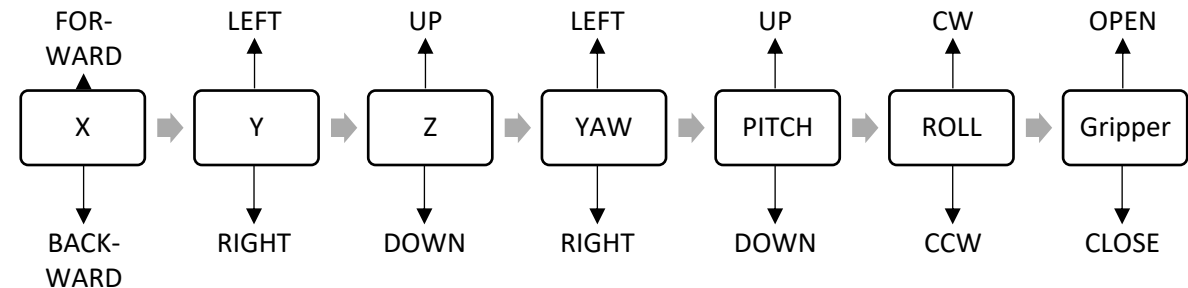


## Cartesian control

笛卡尔空间控制

- 直接控制末端执行器的(x,y,z)位置
- 由机器人控制器完成逆运动学求解
- 移动过程中夹持器姿态可以保持不变

- Control of (x,y,z) end effector position
  - **Inverse kinematics**, done by the ARM controller
- Gripper orientation can be kept unchanged during movement



1. 遥控操作：任务全程由用户直接控制机器人运动。需要多路控制指令，实现三维空间的移动与姿态调整。
  2. 半自主等级：用户按下开关，机器人复现预先存储好的运动轨迹。
- 设计思路：把任务控制权尽可能交给使用者。但是部分场景用户希望机器人独立完成全部动作（例如拾取地面物体）；另一些场景，用户希望保留一部分控制，干预执行过程（比如拿杯子喝水）。

# Levels of Autonomy

3. 完全自主：接收高层任务指令（例如：从架子拿起杯子，接水，送到使用者面前）。机器人自主拆解子任务，自主做决策（例如倒水，判断水是否接满），全程不需要人为干预。

更高等级的自主，需要更强的环境感知、决策能力。机器人需要感知多变的环境，选择最优方案完成任务。

- **Teleoperated:** the user has direct control over the robot movements throughout the task. Multiple controls are then necessary to make the robot move and orient itself in the three-dimensional space.
- **Semi-autonomy levels:** the user merely needs to hit, or press and hold, a switch to replay a prestored movement.  
The rationale is to provide users as much control over the task as they want. While in some cases the user may rely on the robot to perform the task all by itself (e.g., reach out for an object on the floor), there are other cases when the user would like to still keep some of the control to influence how the task is performed (e.g., drinking from a glass).
- **Fully autonomous:** the robot can accept high-level commands that define a task to be completed (e.g., take a glass from the shelf, fill it with water, and bring it to the user). The robot will perform whatever subtasks are necessary and make necessary decisions (e.g., pouring water into the glass and determining the glass is full) without requesting any human intervention.
- Higher levels of robot autonomy require advanced sensing and decision capabilities. The robot needs to be able to sense its environment, which is variable depending on the location and time, and decide on the best approach to carry out a given task.

方框15.1 谢里丹机器人自主等级

1. 计算机不提供任何辅助，全部由人完成。
2. 计算机给出一整套可选行动方案。
3. 计算机将选项精简为少数几个备选。

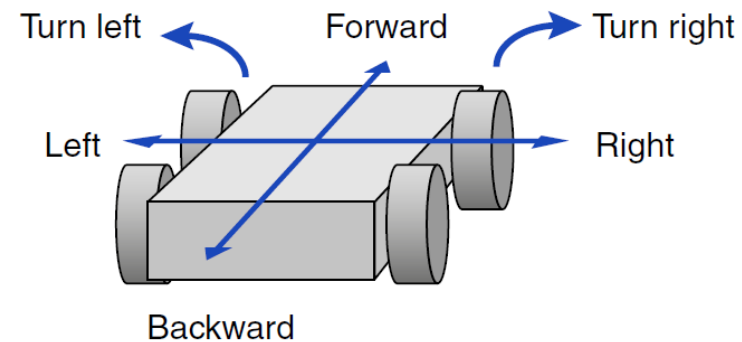
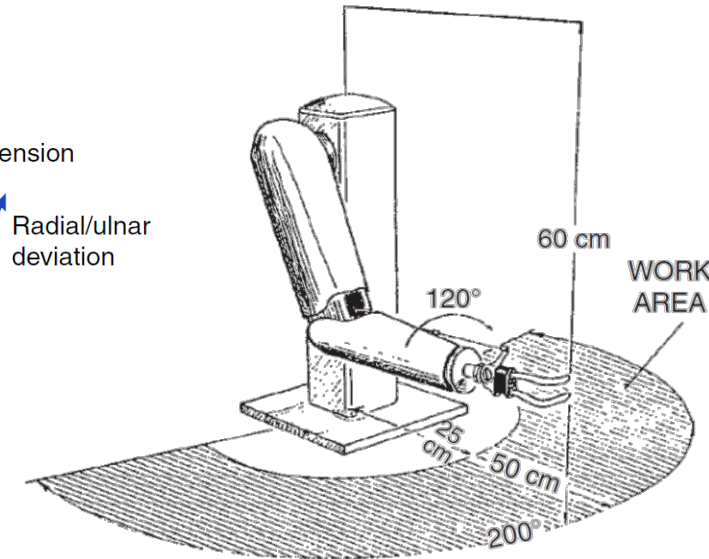
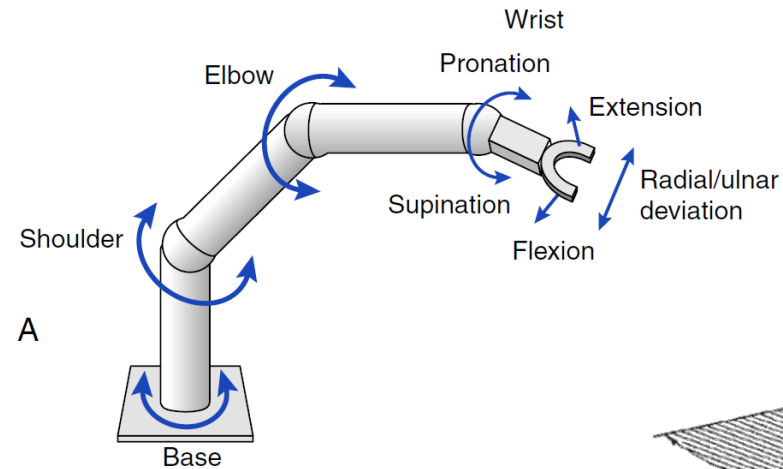
## BOX 15.1 Sheridan's Levels of Robotic Autonomy (Parasuraman et al., 2000)

1. Computer offers no assistance; human does it all.
2. Computer offers a complete set of action alternatives.
3. Computer narrows the selection down to a few choices.
4. Computer suggests a single action.
5. Computer executes that action if human approves.
6. Computer allows the human limited time to veto before automatic execution.
7. Computer executes automatically then necessarily informs the human.
8. Computer informs human after automatic execution only if human asks.
9. Computer informs human after automatic execution only if it decides to.
10. Computer decides everything and acts autonomously, ignoring the human.

4. 计算机推荐单一行动方案。
5. 经人类许可后，计算机执行该行动。
6. 自动执行前，计算机留给人类有限时间来否决该操作。
7. 计算机自动执行，执行完成后必须告知人类。
8. 自动执行完成后，仅当人类主动询问时，计算机才向人类汇报。
9. 自动执行完成后，仅在计算机自身决定有必要时，才告知人类。
10. 计算机自主决定全部事项并独立执行，无视人类。

# ARMs categories

- 机器人工作站：桌面安装 / 吊装在轨道上
- **Robotic workstations**
  - Table-mounted or
  - Attached to overhead tracks
- **Wheelchair-mounted**
  - 轮椅搭载式机械臂
- **Mobile robots**
  - 移动机器人



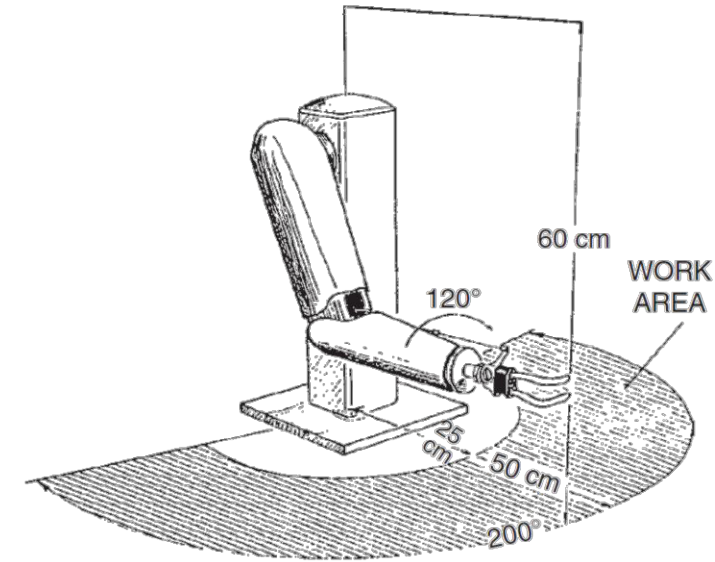
机器人工作站.

# Robotic workstations

APL RAWT

DeVAR/ProVAR

RAID



工作站：专门用于完成特定工作、活动的固定区域。举例：

- 设计工作站（工科学学生电脑工作站）
- 阅读工作站（图书馆工位）
- 文书处理工作站（处理文档、接打电话、整理文件）

# Robotic Workstations

- Workstation: an area dedicated to the performance of a specific job or activity. Examples:
  - design (e.g., a computer workstation for engineering students),
  - reading (e.g., a library-based workstation),
  - clerical tasks (e.g., a workstation for word processing, telephone answering, and manipulation of files).
- These activities involve manipulation of papers, books, and devices.
- Because the workstation is fixed in one location, the design of the robotic system can focus on manipulation of objects in a **structured** environment, without having to deal with uncertainties (e.g., obstacles that may get in the way) characteristic of unstructured environments.

这类活动需要操作纸张、书本、设备。

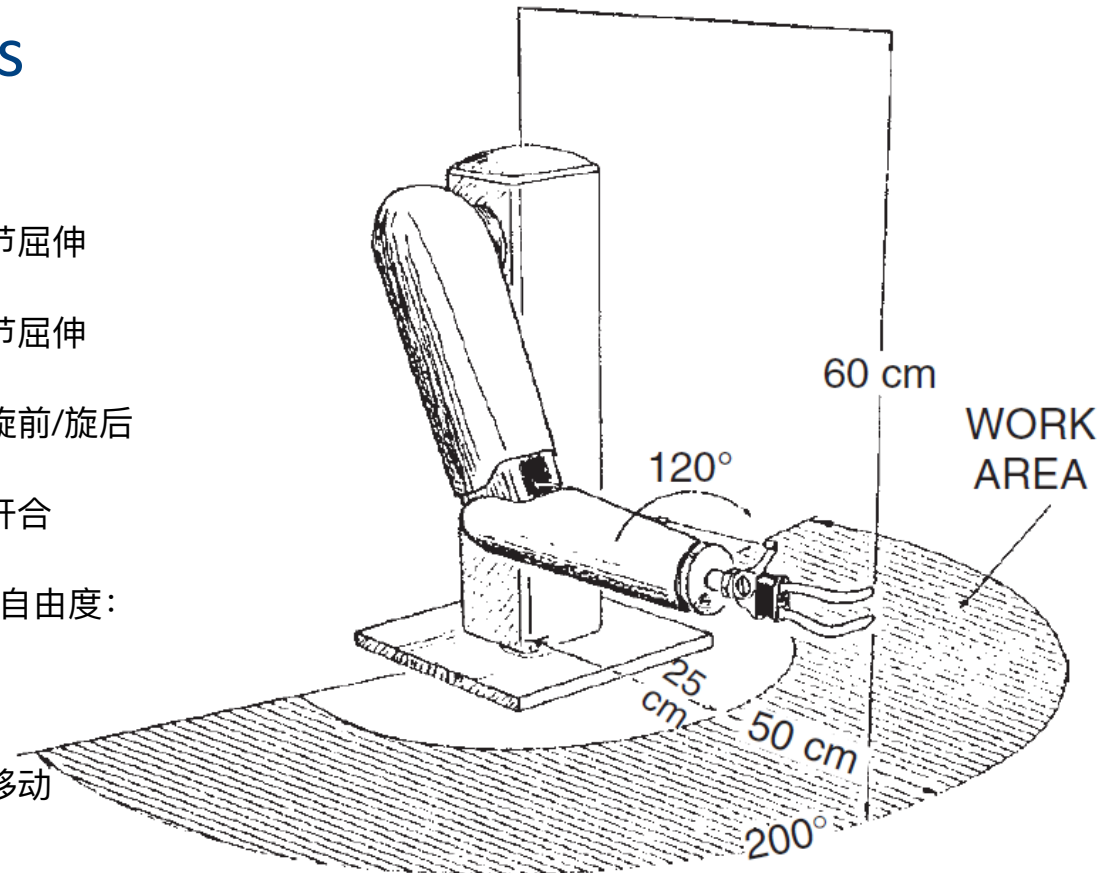
工作站位置固定，机器人系统设计只需要处理该固定区域内物体操作；不需要处理非结构化环境中各类不确定因素，例如随机障碍物。

# APL RAWT

APL RAWT (应用物理实验室机器人臂工作台)

## (Applied Physics Laboratory Robot Arm WorkTable)

- Built from an upper limb prosthesis
- 4 DOF: 由上肢假肢改造而来  
4个臂部自由度:
  1. Shoulder flexion/extension 1. 肩关节屈伸
  2. Elbow flexion/extension 2. 肘关节屈伸
  3. Wrist pronation/supination 3. 腕部旋前/旋后
  4. Hand opening/closing 4. 手部开合
- 2 DOF in the base:
  1. rotation 底座2个自由度:  
1. 旋转
  2. forward/backward 2. 前后移动



DeVar / ProVar

加州帕洛阿尔托美国退伍军人事务部研发中心

# DeVar/ProVar

## VA R&D Rehabilitation Center, Palo Alto, CA

- A PUMA robot used in a workstation
- Ottobock hand prosthesis as end effector

- 将PUMA工业机器人用于工作站
- 末端执行器采用奥托博克假肢手  
初代原型机：1982年



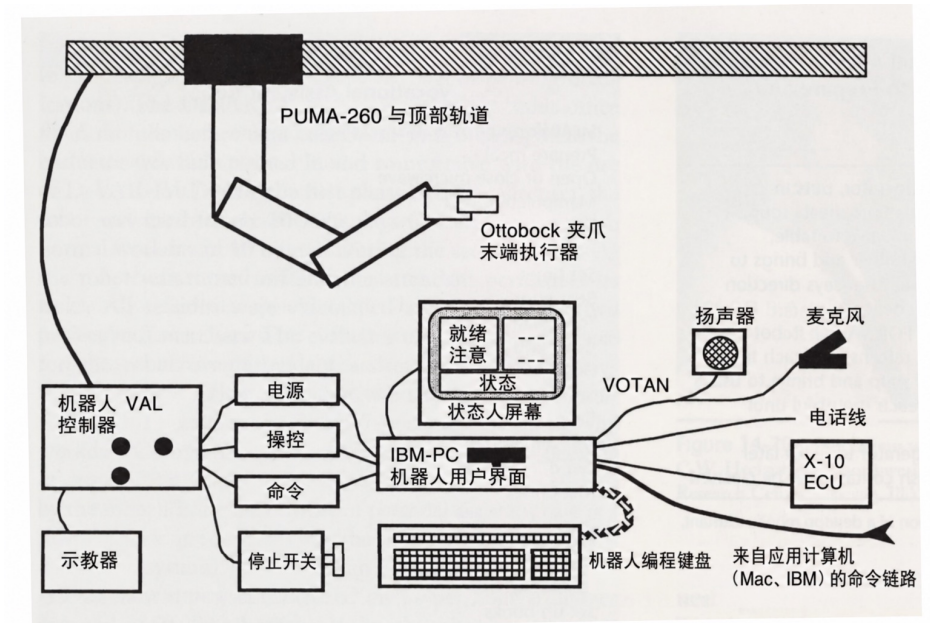
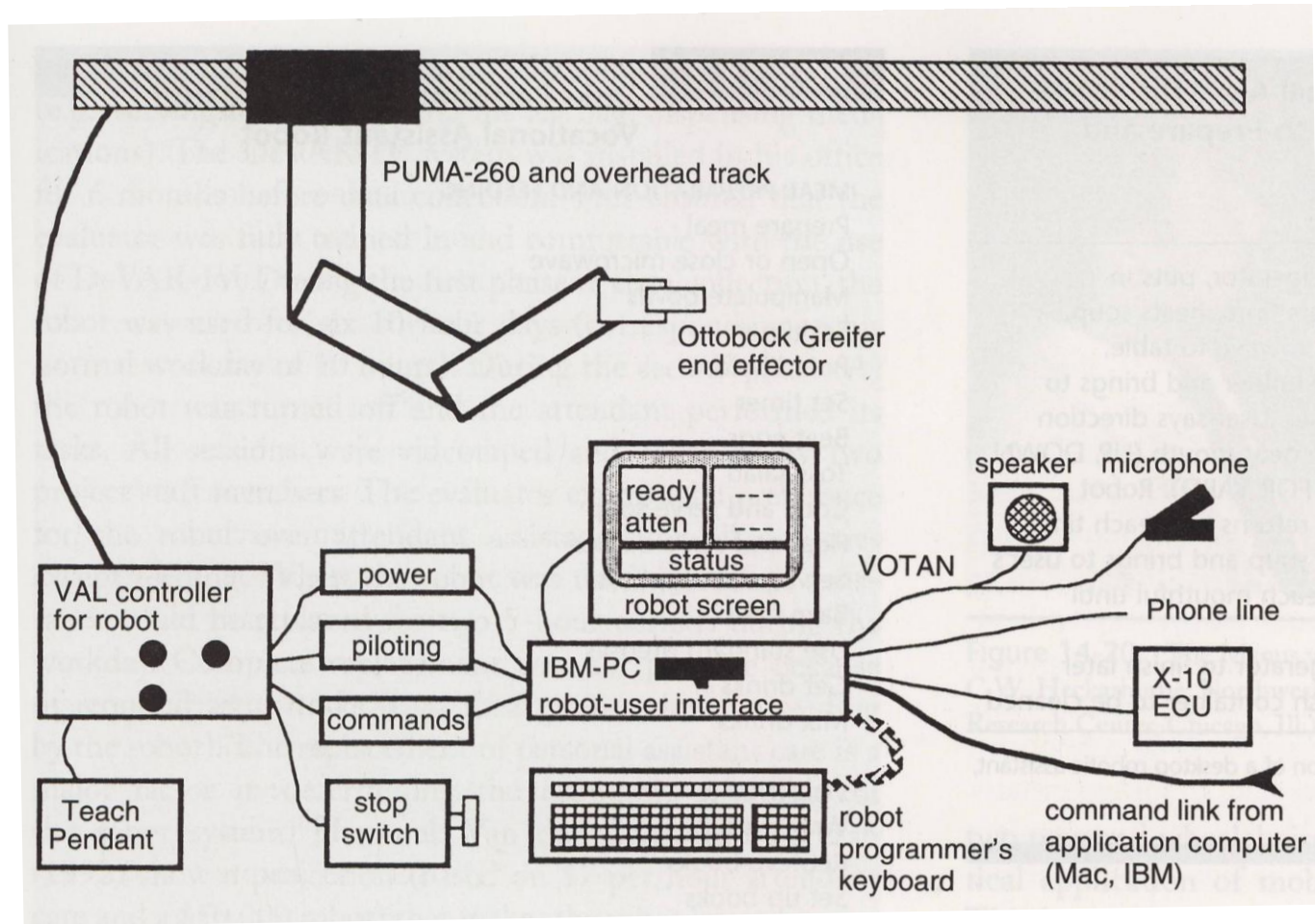
*First prototype: 1982*

# DeVar control

DeVar控制系统

加州帕洛阿尔托美国退伍军人事务部研发中心

## VA R&D Rehabilitation Center, Palo Alto, CA



First prototype: 1982

# MASTER/RAID

- The user was able to control a robot mounted on a horizontal rail that ran in front of a metallic structure with shelves
- A vertical axis enabled the robot to access shelves at different heights, and a gripper mechanism provided a means to retrieve objects from the shelves
- These objects could then be placed on the user's desk

机器人安装在水平导轨上，导轨前方布置带多层隔板的金属置物架。竖直方向轴，让机械臂可以到达不同高度的隔板；夹持机构从架子取出物品，放到用户桌面上。

初代原型机：1993年



*First prototype: 1993*

轮椅搭载机械臂.

# Wheelchair-mounted robotic arms

Manus/iARM

JACO

# Manus / iARM

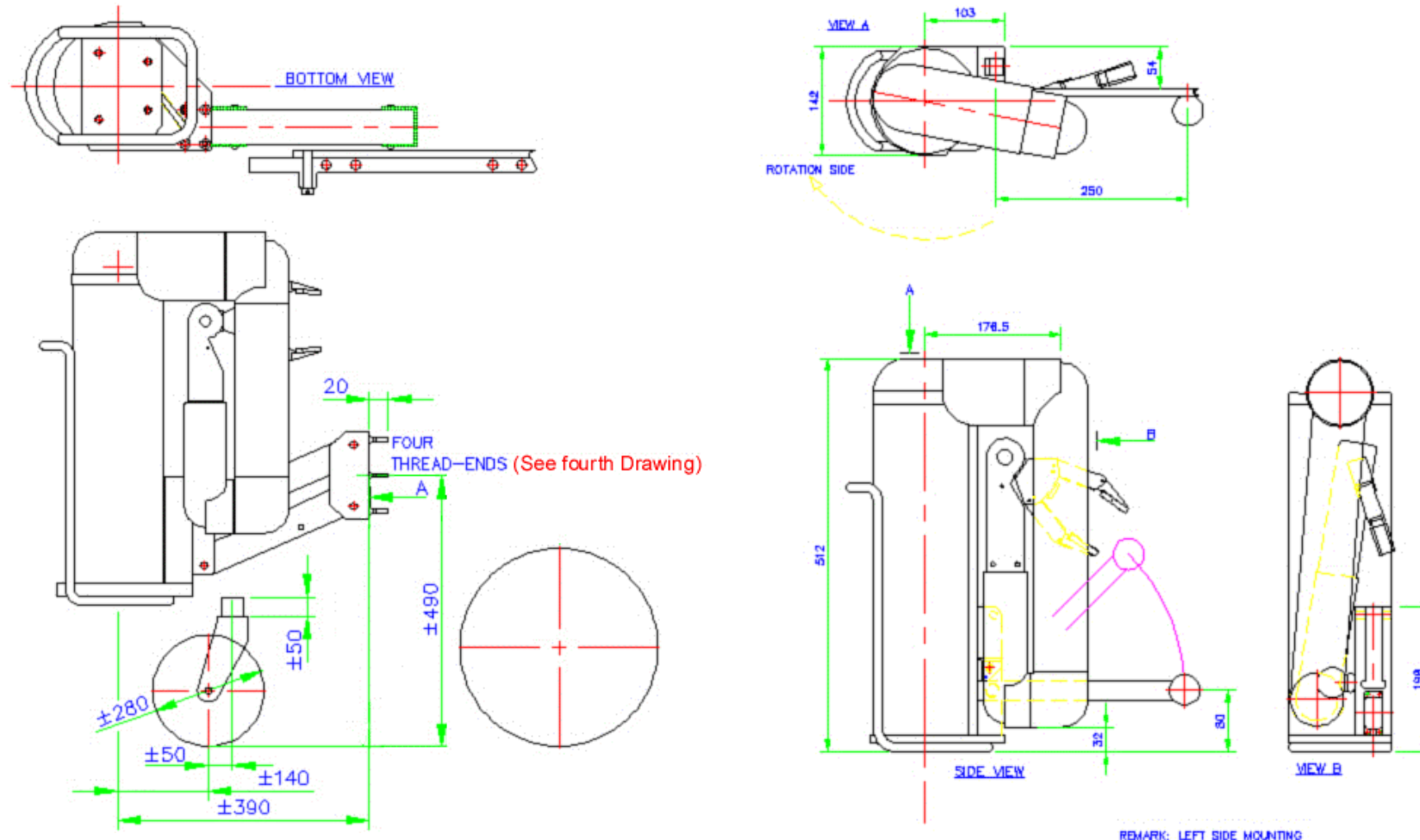
荷兰Exact Dynamics公司 Manus / iARM

## by Exact Dynamics, The Netherlands

- 8 DOF
  - 8自由度
  - 重量：9千克
  - 完全伸展下有效负载：1.5千克
  - 夹持力：20牛
  - 可反向驱动的皮带传动
  - 直接使用轮椅电池供电
- Weight: 9 Kg
- Payload: 1.5 Kg  
when fully extended
- Gripping force: 20 N
- Backdrivable belt drive
- Uses wheelchair battery as  
source of power



# Manus / iARM geometry and kinematics



# Manus / iARM control

支持多种控制：

- It can be controlled using
  - direct selection through a keypad or a joystick or
  - indirect selection (scanning) through a single switch
- It allows Cartesian control, enabling the user to move the robot gripper up/down, left/right, and forward/backward while the robot automatically adjusts each joint to keep a cup or plate level during movement.
- It stores particular positions that are frequently used. They can be recalled with one selection
  - 支持笛卡尔空间控制：机械臂末端可以上下、左右、前后移动；机器人自动调节各个关节，保证杯子、餐盘保持水平。
  - 可以存储常用位置，菜单一键调用。



# JACO by Kinova, Canada

- 7 DOF
  - 3-DOF wrist: pronation/supination, flexion/extension, radial/ulnar deviation
- Equipped with a two-finger or a three-finger hand
  - 1-DOF hand: open and close
- The end effector can be moved in 3D space
  - up/down
  - left/right
  - forward/backward
- Maximum reach 900 mm
- Weight: 5.2 Kg
- Payload: 1.6 Kg
- Lighter 6-DOF version available (4.4 kg)

- 7自由度
- 腕部3自由度：旋前旋后、屈伸、桡尺侧偏移
- 二指或者三指机械手
- 手部1个自由度：张开、闭合
- 末端执行器可以在三维空间移动：上下 / 左右 / 前后
- 最大工作伸展距离：900毫米
- 整机重量：5.2千克；6自由度轻量化版本重4.4千克
- 负载：1.6千克



# SPECIFICATIONS



|                                 |                                   |
|---------------------------------|-----------------------------------|
| Weight                          | 5.2 kg / 11.5 lbs                 |
| Payload (full-range continuous) | 1.3 kg / 2.9 lbs                  |
| Payload (mid-range continuous)  | 1.6 kg / 3.5 lbs                  |
| Maximum reach                   | 900 mm / 2.95 ft                  |
| Materials                       | Carbon fiber and aluminum         |
| Maximum speed                   | 20 cm/s                           |
| Power supply voltage            | 18 to 29 VDC                      |
| Average power                   | 25 W (5 W on standby mode)        |
| Peak power                      | 100 W                             |
| Ingress protection              | IPX2                              |
| Operating temperature           | -10 °C to 40 °C / 14 °F to 104 °F |



Reach us

<https://assistive.kinovarobotics.com/product/jaco-robotic-arm>



# JACO control

通过2个按键切换3种工作模式：

- 3 control modes, accessed by 2 push buttons
  1. **Translation mode:** the user can move the robot's hand in 3D space with the orientation of the hand fixed. 1. 平移模式：机械手在三维空间移动，手部姿态保持不变。
  2. **Wrist mode:** the user can modify the orientation of the hand, but the hand is kept centered at the same point in space. 2. 腕部模式：手部空间位置固定不变，只改变手部姿态。
  3. **Finger mode:** the user can grasp and release the hand, using either 2 or 3 fingers. 3. 手指模式：机械手张开、闭合，支持二指/三指版本。

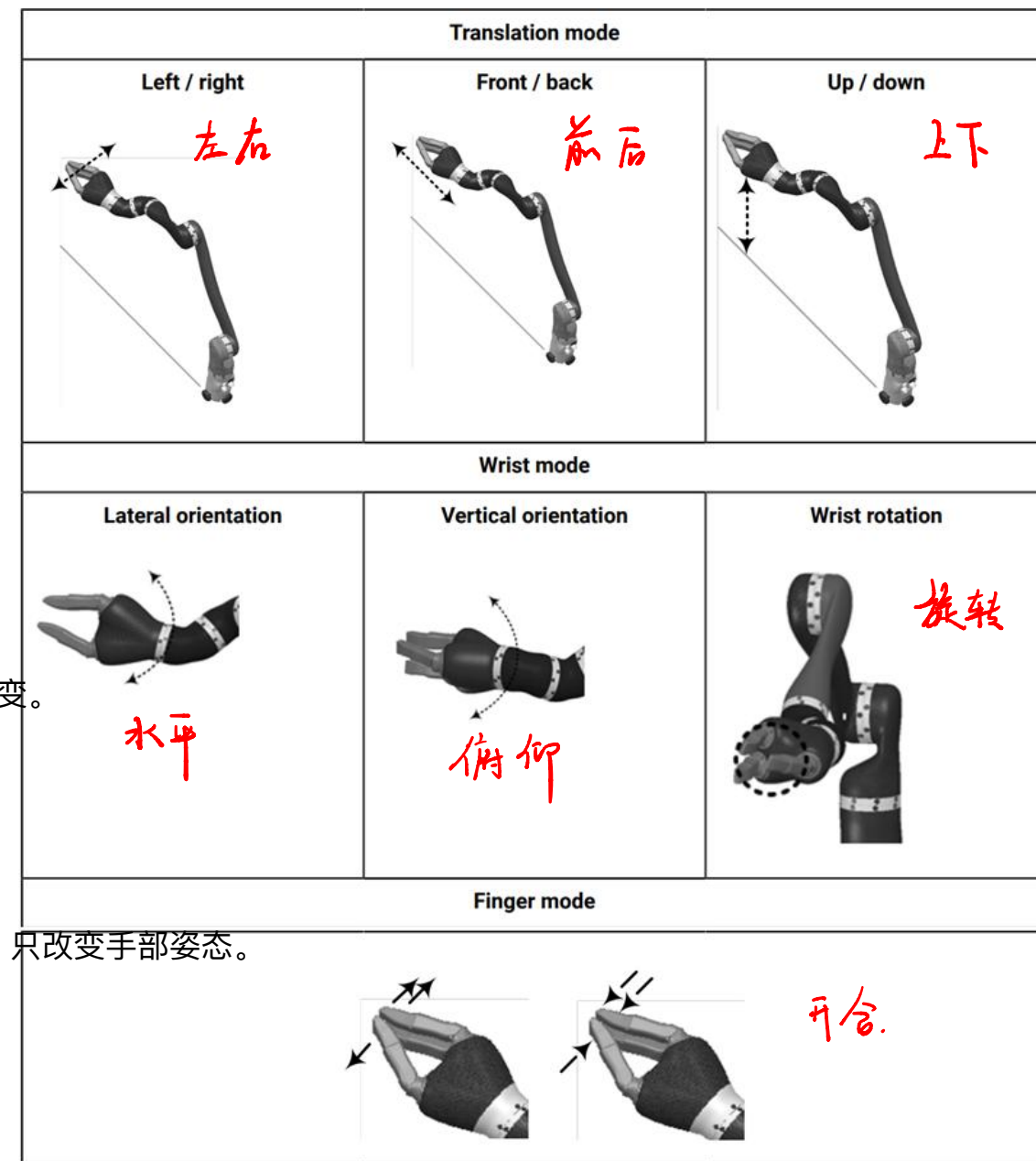


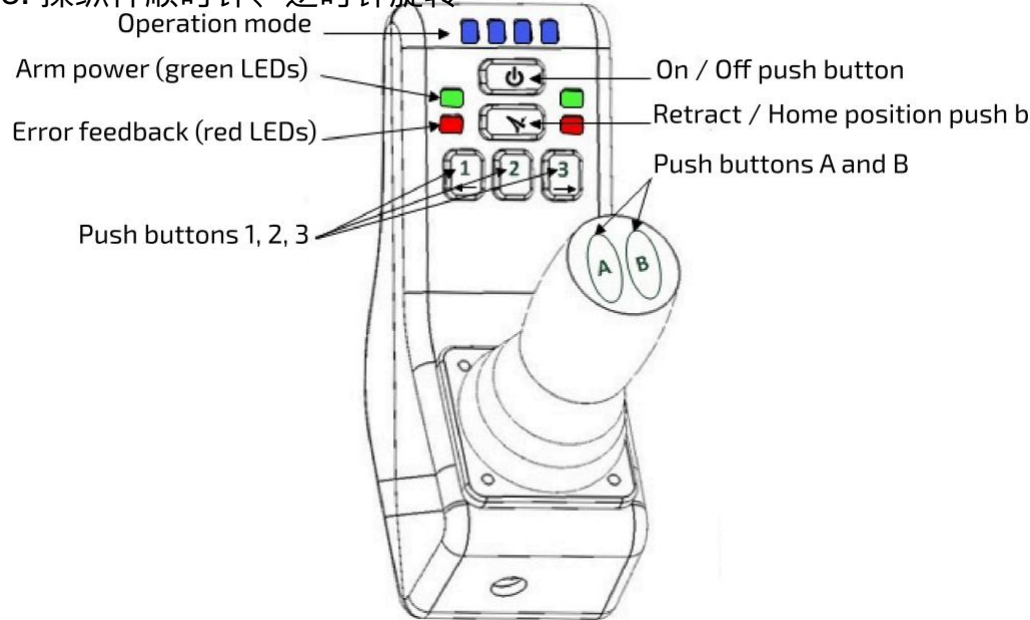
Table 4: Three-axis mode joystick controls

# JACO control

JACO配备三轴操纵杆，安装在轮椅扶手：

- 3-axis joystick, mounted on the wheelchair armrest
  1. shaft forward or backward
  2. shaft right or left
  3. handle turned clockwise and counter clockwise

1. 操纵杆前后推
2. 操纵杆左右拨动
3. 操纵杆顺时针、逆时针旋转



<https://assistive.kinovarobotics.com/uploads/EN-UG-007-Jaco-Assistive-robot-user-guide-r06.1.pdf>

| Joystick movement             | Robot movement                     | Availability        |
|-------------------------------|------------------------------------|---------------------|
| Translation mode              |                                    |                     |
| Incline FRONT                 | Gripper moves forward              | 4 / 6 / 6S / 7S DoF |
| Incline BACK                  | Gripper moves backward             | 4 / 6 / 6S / 7S DoF |
| Incline LEFT                  | Gripper moves left                 | 4 / 6 / 6S / 7S DoF |
| Incline RIGHT                 | Gripper moves right                | 4 / 6 / 6S / 7S DoF |
| Rotate stick CLOCKWISE        | Gripper moves up                   | 4 / 6 / 6S / 7S DoF |
| Rotate stick COUNTERCLOCKWISE | Gripper moves down                 | 4 / 6 / 6S / 7S DoF |
| Wrist mode                    |                                    |                     |
| Incline FRONT                 | Vertical orientation – top side    | 6 / 6S / 7S DoF     |
| Incline BACK                  | Vertical orientation – bottom side | 6 / 6S / 7S DoF     |
| Incline LEFT                  | Lateral orientation – thumb side   | 6 / 6S / 7S DoF     |
| Incline RIGHT                 | Lateral orientation – index side   | 6 / 6S / 7S DoF     |
| Rotate stick CLOCKWISE        | Wrist rotation clockwise           | 4 / 6 / 6S / 7S DoF |
| Rotate stick COUNTERCLOCKWISE | Wrist rotation counterclockwise    | 4 / 6 / 6S / 7S DoF |
| Finger mode                   |                                    |                     |
| Incline LEFT                  | Close Fingers (3-finger mode)      | 4 / 6 / 6S / 7S DoF |
| Incline RIGHT                 | Open Fingers (3-finger mode)       | 4 / 6 / 6S / 7S DoF |

# JACO – iARM comparison

Characteristics of the Commercially Available JACO and iARM

商用JACO、iARM对比表

| Characteristic           | iARM                                                                                                           | JACO                                                                                                         |
|--------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Manufacturer             | Exact Dynamics                                                                                                 | Kinova Technology                                                                                            |
| Weight                   | 9 kg                                                                                                           | 5 kg                                                                                                         |
| Weight limit             | 1.5 kg                                                                                                         | 1.5 kg (45 cm), 1.0 kg (90 cm)                                                                               |
| Reach                    | 90 + 20 cm (lift unit)                                                                                         | 90 cm                                                                                                        |
| Maximum speed            | 15 cm/s                                                                                                        | 15 cm/s                                                                                                      |
| Degrees of freedom (DOF) | 7, including gripper; 8 with lift unit                                                                         | 7, including gripper                                                                                         |
| Hand                     | 2 fingers                                                                                                      | 3 fingers                                                                                                    |
| Finger force             | 20 N                                                                                                           | 7 N                                                                                                          |
| Control possibilities    | Several power wheelchair interfaces, keypad, joystick (digital or analog), single switches, game pad, computer | Several power wheelchair interfaces, mini-joystick, three-axis joystick, button controls, game pad, computer |
| Power                    | 24-V DC/3 A (maximum)                                                                                          | 24-V DC/1.5 A                                                                                                |

|        |                        |                                 |
|--------|------------------------|---------------------------------|
| 特性     | iARM                   | JACO                            |
| 制造商    | Exact Dynamics         | Kinova                          |
| 重量     | 9kg                    | 5.2kg                           |
| 最大伸展距离 | 90cm+提升单元20cm          | 90cm                            |
| 负载     | 1.5kg                  | 1.5kg (45cm) , 1.0kg (90cm)     |
| 最大速度   | 15cm/s                 | 15cm/s                          |
| 自由度    | 7 (含夹持器) , 带提升单元共8     | 7 (含夹持器)                        |
| 手部夹持力  | 20N, 二指                | 7N, 三指                          |
| 控制接口   | 电动轮椅接口、键盘、数字/模拟操纵杆、单开关 | 电动轮椅接口、迷你操纵杆、三轴操纵杆、按键、游戏手柄、电脑控制 |
| 供电     | 24V直流 /最大3A            | 24V直流 /最大1.5A                   |

Source: Adapted from Chung, C. S., and Cooper, R. A. 2012. "Literature review of wheelchair-mounted robotic manipulation: user interface and end-user evaluation." Proceedings of the 12th Annual RESNA Conference, Baltimore.

可移动机器人

# Mobile robots

MOVAID

Care-o-Bot

作为辅助设备的机器人

# Mobile robots as assistive devices

轮式移动底盘搭载机械臂，配有简易置物托盘搬运物体，也可以实现环境设备控制。

- Wheeled mobile robots equipped with an arm and a simple tray to carry objects or capable of controlling the environment
- Common features:
  - monitoring the user, including video-monitoring by a caregiver or health care provider
    - 用户状态监控（护理人员远程视频查看）
  - helping the user to accomplish activities of daily living, either by providing
    - 辅助用户完成日常生活活动：
      - cognitive support (e.g., warning the person that it is time to take a pill or prompting the person through the steps of cooking a meal)    ◦ 认知辅助（例如提醒吃药，一步步引导烹饪）
      - physical support (e.g., retrieving a glass of water)    ◦ 物理操作辅助（拿一杯水）

# MOVAID

MOVAID (残障人群移动与活动辅助系统)

## (Mobility and Activity Assistance Systems for the Disabled)

- 2-DOF mobile base
  - 移动底盘2自由度
- 8-DOF robotic arm
  - 机械臂8自由度
- 2-DOF 3-finger hand
  - 三指机械手, 2自由度
- 2-camera vision system
  - 双摄像头视觉系统
- Laser-based localization
  - 激光定位
- US obstacle avoidance
  - 超声波避障
- Tray for object transportation
  - 物品搬运托盘
- Docking system
  - 对接机构, 可以对接家庭内固定工作站
- Fixed workstations in the home environment



欧盟TIDE项目, 比萨圣安娜高等学校, 1994-1997

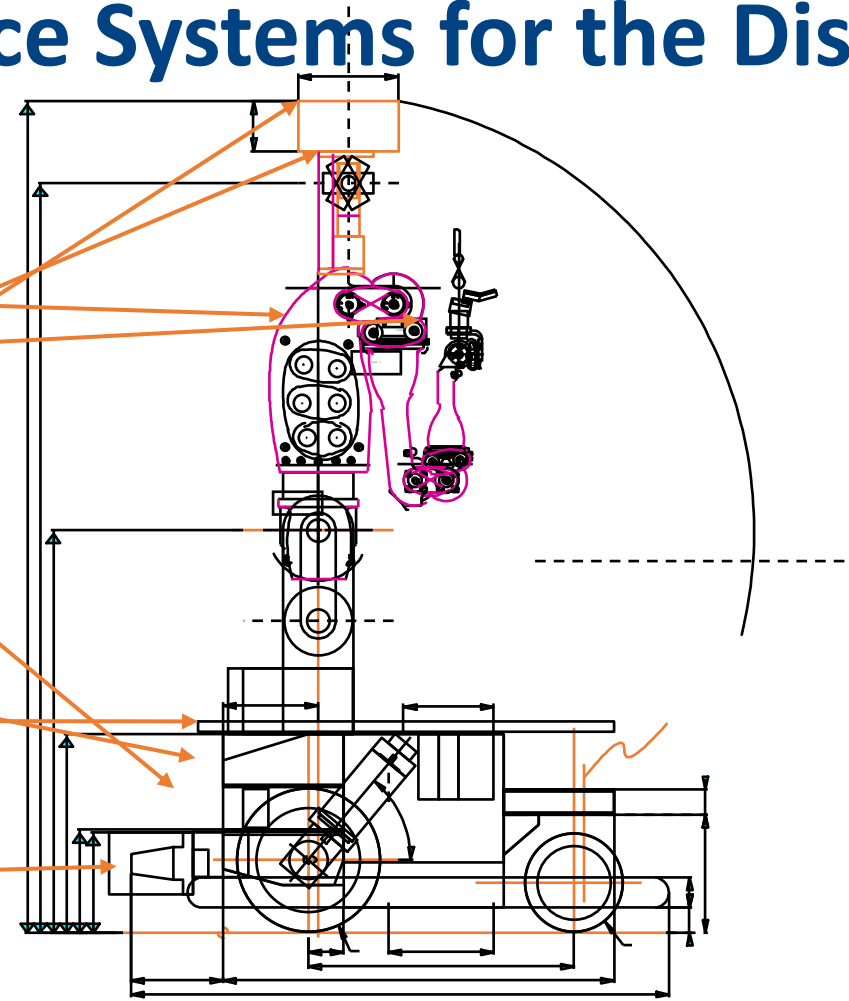
EC TIDE project, Scuola Superiore Sant'Anna, Pisa, 1994-1997

Dario P., Guglielmelli E., Laschi C., Teti G.(1999) "MOVAID: A personal robot in everyday life of disabled and elderly people" *Technology and Disability* 10(2):77-93.

# MOVAID

## (Mobility and Activity Assistance Systems for the Disabled)

- 2-DOF mobile base
- 8-DOF robotic arm
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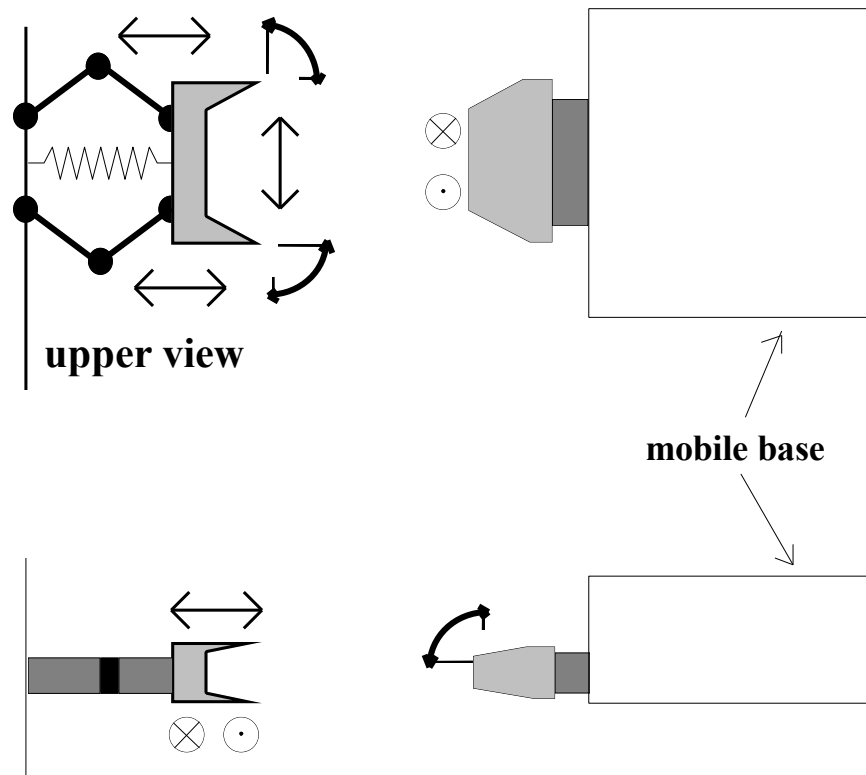


*EC TIDE project, Scuola Superiore Sant'Anna, Pisa, 1994-1997*

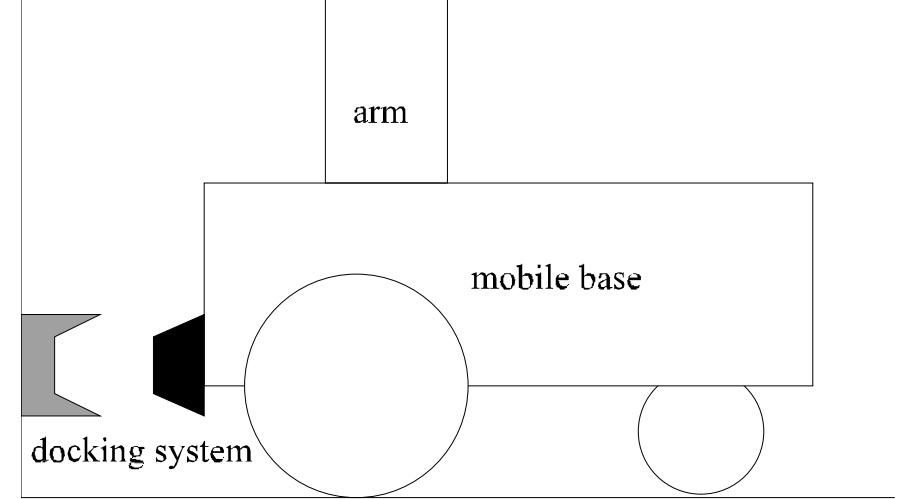
MOVAID对接功能

# MOVAID docking

对接完成之后，机器人运行在结构化环境，工作模式和固定机器人工作站一致。  
**When docked, the robot operates in a structured environment, similarly to a robotic workstation**



side view



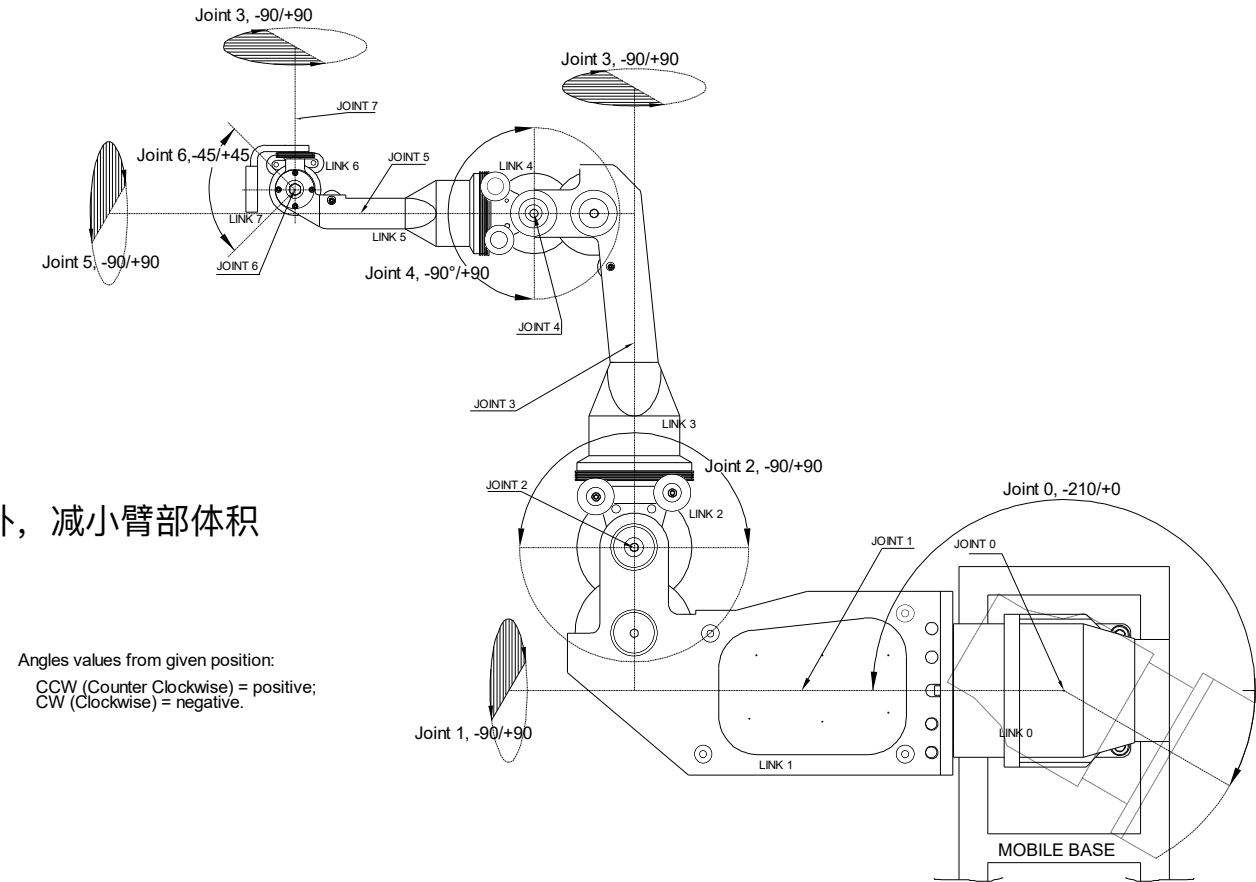
# Dexter arm by Scientia Machinale, Italy

- 8自由度
- 最大伸展距离: 1500毫米
- 重复定位精度:  $\pm 1$ 毫米
- 最大速度: 0.2米/秒
- 8 DOF
- Maximum reach: 1500 mm
- Repeatability: + 1mm
- Maximum speed: 0.2 m/s
- Weight: 40 Kg
- Payload: 2 Kg
- Cable-driven: reduced size, since motors are outside the arm

重量: 40千克

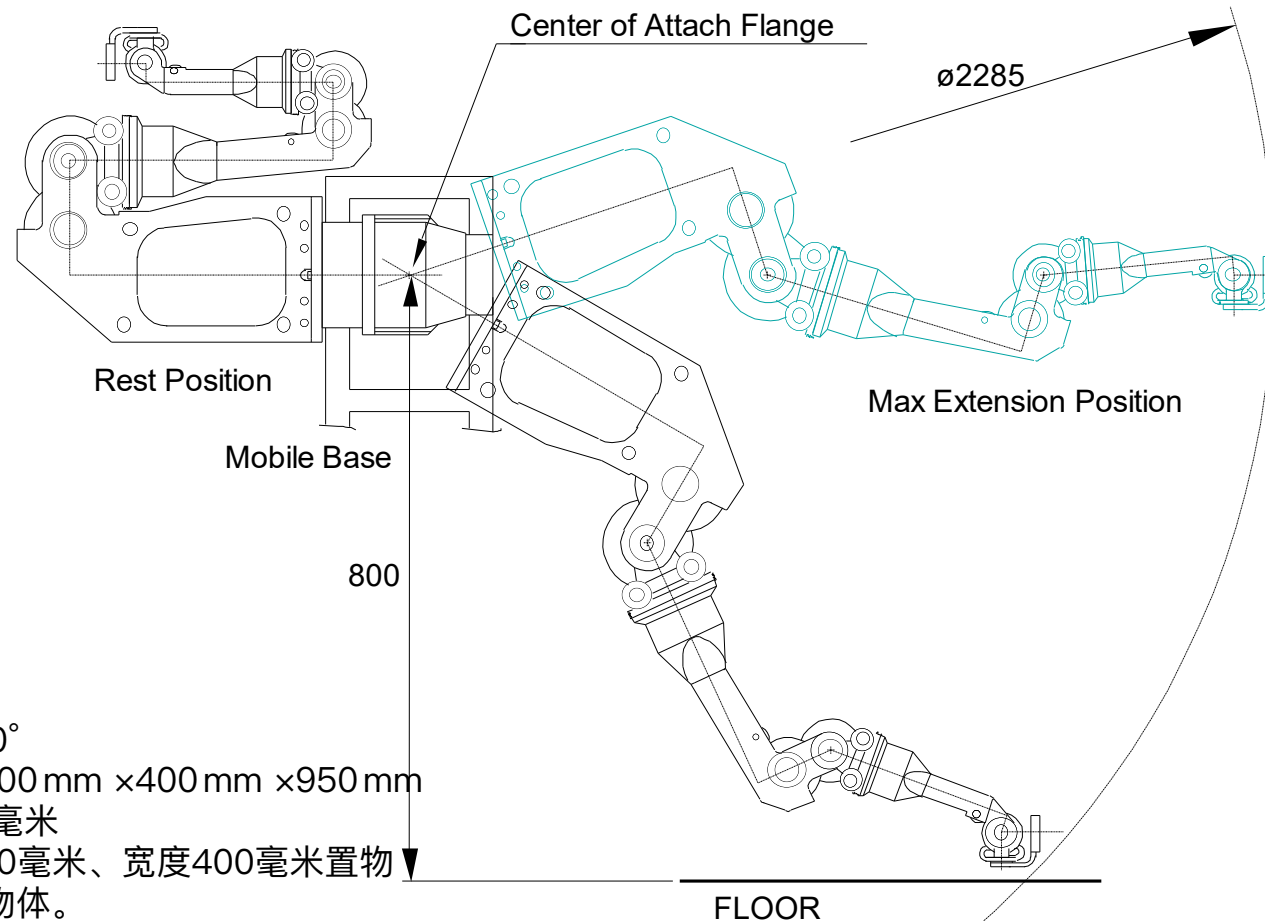
• 负载: 2千克

• 绳索驱动: 电机布置在臂体外, 减小臂部体积



# Dexter arm by Scientia Machinale, Italy

- Workspace: 1200 mm x 350°
  - Foldable structure:
    - 400 mm x 400 mm x 950 mm
  - Sagittal plane workspace is a circle of diameter 2285 mm
  - The arm can reach an object
    - on a 400-mm wide shelf at 1500 mm height
    - on the floor
- 工作空间: 1200毫米 x 350°  
 • 可折叠结构; 折叠尺寸: 400 mm x 400 mm x 950 mm  
 • 矢状面工作范围直径2285毫米  
 • 机械臂可以够取: 高度1500毫米、宽度400毫米置物架上物体, 也可以够取地面物体。

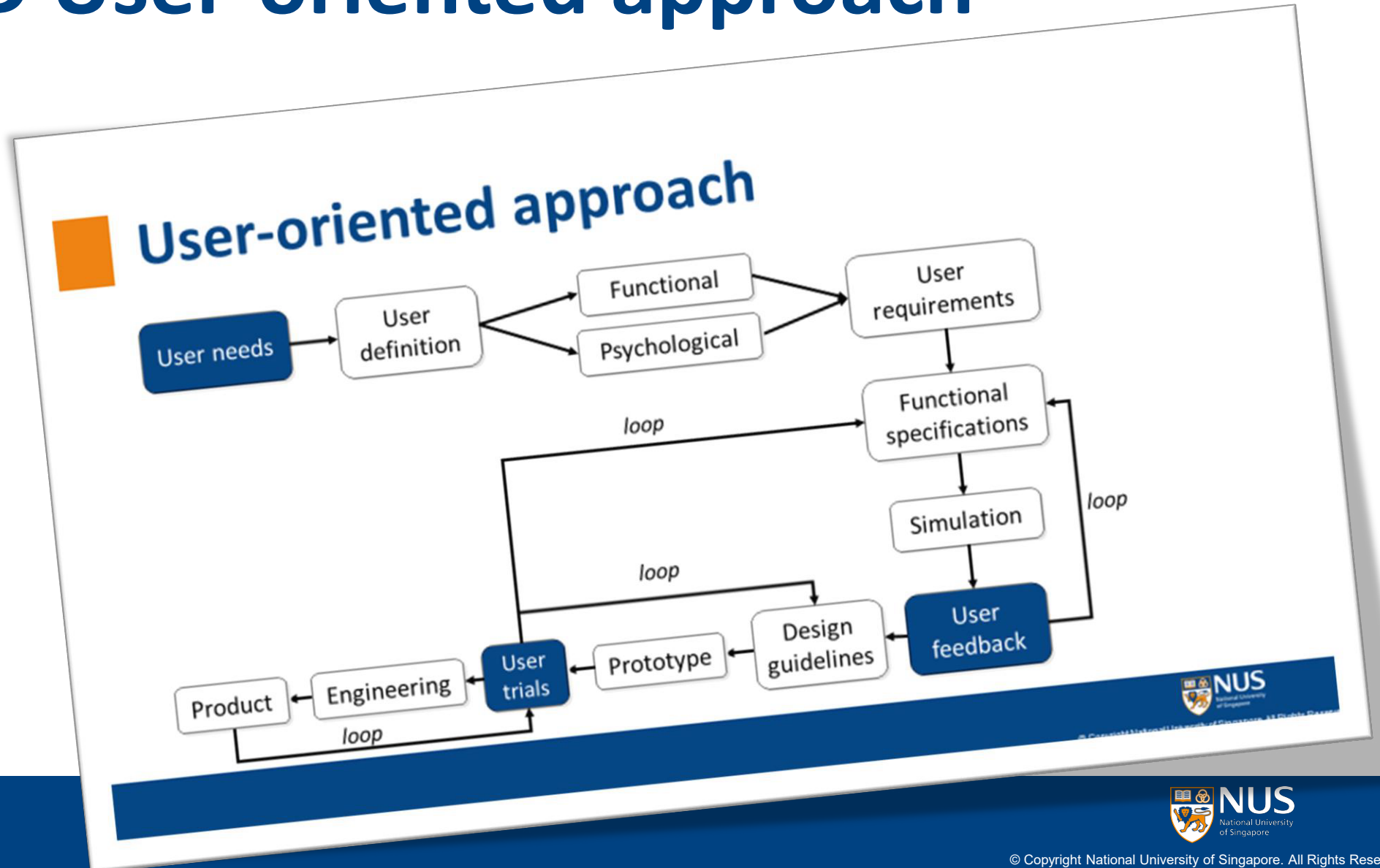


MOVAID

面向用户的设计流程

# MOVAID User-oriented approach

From intro  
lecture

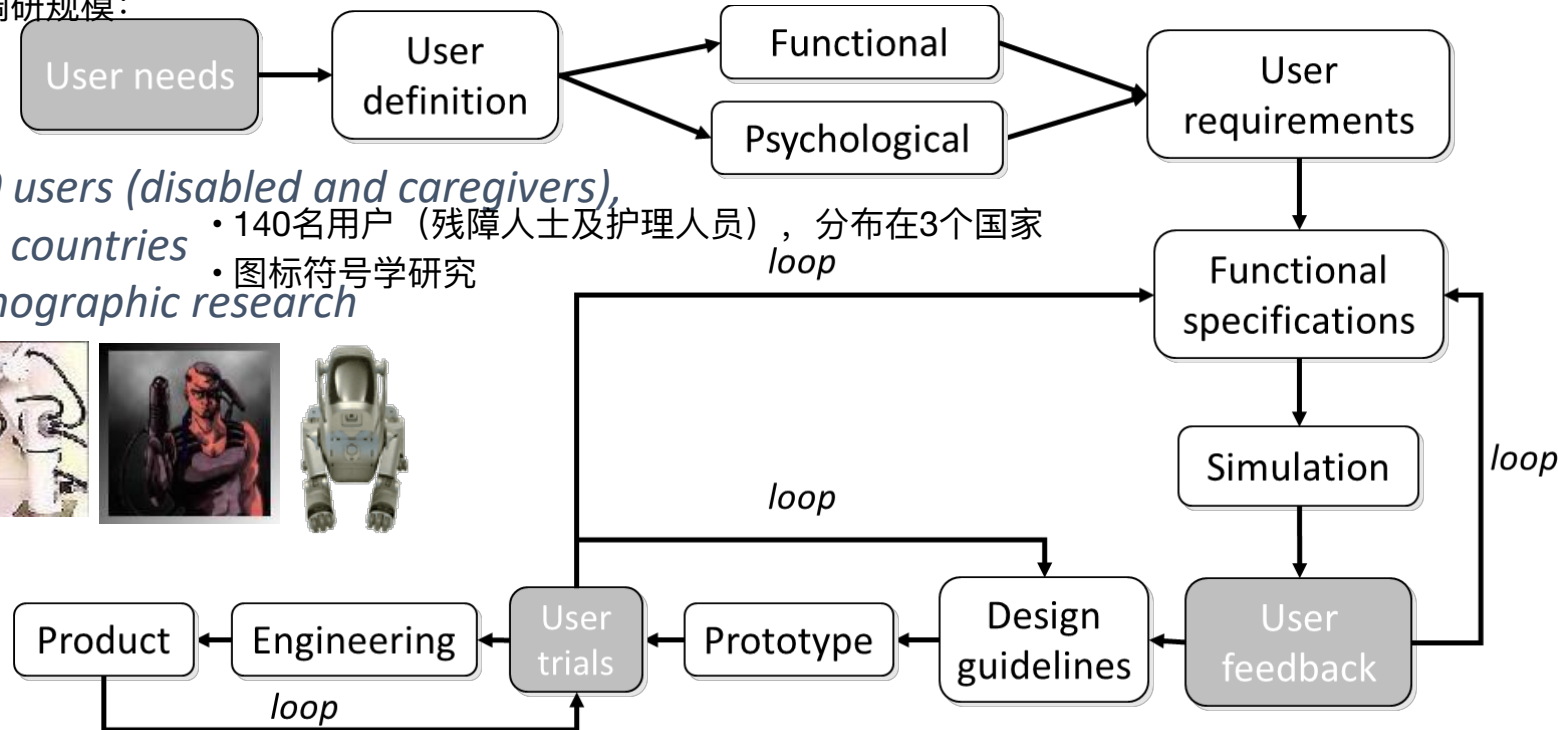


# MOVAID User-oriented approach

用户调研规模：

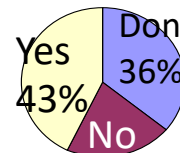
*140 users (disabled and caregivers),  
in 3 countries*  
*Iconographic research*

- 140名用户（残障人士及护理人员），分布在3个国家
- 图标符号学研究



*70 users, 7 validation sites (hospitals,  
rehabilitation centres) in 3 countries:  
Italy, France, Switzerland*

Would you like an assistive robot?



调研问题：你是否需要一台辅助机器人？

- 是：43%
- 否：21%
- 不确定：36%

*3 demo tasks*

- *Dish warm-up*
- *Kitchen countertop cleaning*
- *Bed sheet removal*

*Semi-autonomy*



- 70名用户、7个验证站点（医院、康复中心），分布在意大利、法国、瑞士三国

3个演示任务（半自主模式下测试）：

1. 餐盘加热
2. 厨房台面清洁
3. 撤换床单

根据用户分析得出的个人机器人理想特征

# Desirable characteristics of a personal robot according to users analysis

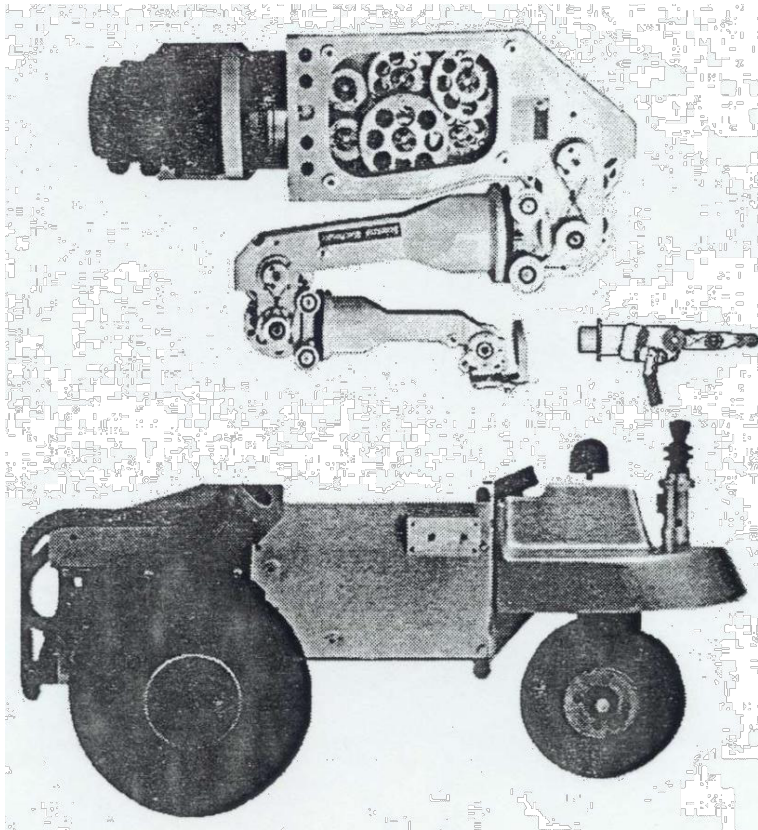
- 机器人外观应明显是人造物，绝不能仿生或拟人化；
- the robot should look clearly **artificial**, never biomorphic nor anthropomorphic;   
• 技术感外观与"柔和"元素之间需谨慎平衡，后者应传递人文关怀与友好感；
- a careful balance should be kept between the technical look and some 'softer' elements suggesting a feeling of humanism and **friendliness**;
- 'caricature' elements (pet-robot, puppet) should be generally avoided, as they could create a sense of unreliability (should I be helped by a toy?);
- the robot should 'sleep' in a **reassuring position** and, when waking up, it should **warn** and start with a **gentle movement**;   
• 机器人应在令人安心的位置"休眠"，启动时应先发出提示，再以平缓的动作开始工作；
- the robot should never move in the darkness, or should **warn with a light** before moving;   
• 机器人绝不能在黑暗中移动，若要移动应先亮灯警示；
- the arm should possibly offer objects from a **lower position**, avoiding coming toward the user with a 'threatening' attitude.

• 机械臂应尽量从较低位置向用户递物，避免以"威胁性"姿态朝向用户移动。

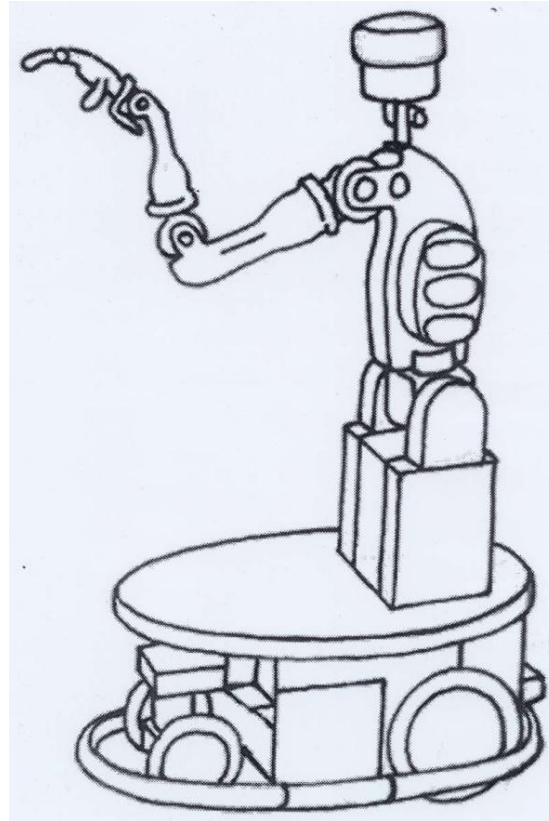
*By industrial designers*

# MOVAID industrial design

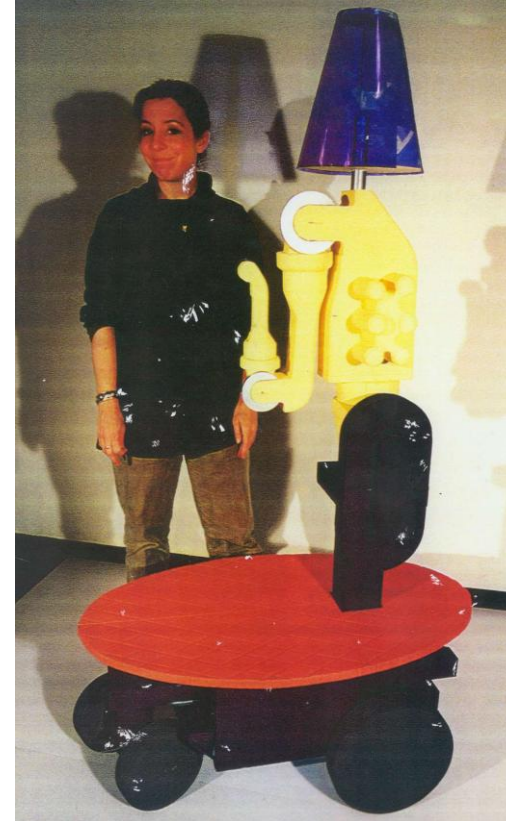
零部件选型 → 美学设计 → 木质样机 → 原型机



Starting components



Aesthetic design



Wooden mock-up



Prototype

MOVAID 三项任务

# MOVAID tasks

## 1. Dish warm-up

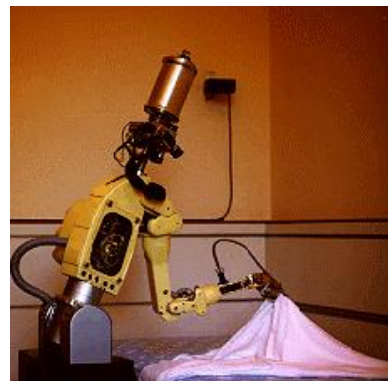
1. 餐盘加热

## 2. Kitchen countertop cleaning

2. 厨房台面清洁

## 3. Bed sheet removal

3. 撤换床单



MOVAID 半自主：人机协作模式

# MOVAID semi-autonomy

- 用户辅助机器人完成对用户简单、对机器人困难的任务，例如：物体识别
- 机器人辅助用户完成操作类任务

## User-robot cooperation

- The user helps the robot in tasks easy for the user and difficult for the robot; e.g., object recognition
- The robot helps the user in manipulation task

工作流程：

1. 用户在界面上点击目标物体

User click

Object image

2. 视觉系统提取物体图像坐标  $(u, v)$

$(u, v)$

Robot controller:

- 3D object localization  $(x, y, z)$
- motion planning  $(x, y, z, \rho, \theta, \varphi)$

$(x, y, z, \rho, \theta, \varphi)$

3. 机器人控制器进行三维物体定位  $(x, y, z)$

4. 运动规划，生成完整位姿  $(x, y, z, \phi, \theta, \psi)$

User interface



# Care-o-Bot

- Household robot that assists people in performing their everyday life at home, such as cleaning table and clearing dish washer.
- Not limited to assisting older people
  - 家用辅助机器人，协助人们在家中完成日常活动，例如：擦桌子、清空洗碗机
  - 不局限于辅助老年人



## TECHNICAL DATA SHEET



Care-O-bot 4



| MODULE                                                                                | COMPONENT/<br>PROPERTY         | VALUE                                                                                                               |
|---------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>FULLY CONFIGURED ROBOT</b>                                                         |                                |                                                                                                                     |
|    | Dimensions (L/W/H):            | 72/72/158 cm                                                                                                        |
|                                                                                       | Weight:                        | 140 kg                                                                                                              |
|                                                                                       | Power Supply:                  | Li-Ion battery 48V<br>External charger 48V@10A<br>Internal levels 48V, 24V, 19V (battery voltage can vary 44V-56V)  |
|                                                                                       | Degrees of freedom:<br>Safety: | 29<br>3 safety laser scanners with safety PLC<br>2-channel Safe-Torque-Off (STO) separate for mobile base and torso |
|    | Communication:                 | 2-channel EM-Stop prepared in all modules<br>2-channel wireless EM-stop                                             |
|                                                                                       | Computing:                     | Gigabit Ethernet in all modules                                                                                     |
|                                                                                       |                                | 4-6 Intel NUC i5, 256GB, 8GB RAM                                                                                    |
|                                                                                       |                                |                                                                                                                     |
| <b>OMNIDIRECTIONAL BASE</b>                                                           |                                |                                                                                                                     |
|    | Dimensions (L/W/H):            | 72/72/33 cm                                                                                                         |
|                                                                                       | Weight:                        | 42 kg                                                                                                               |
|                                                                                       | Power Supply:                  | Li-Ion battery 48 V<br>External charger 48V@10A<br>Internal levels 48V, 24V, 19V                                    |
|                                                                                       | Communication:                 | Gigabit Ethernet / Dual-channel router with DDWRT (2,4GHz+5GHz) / Low-level: CANOpen                                |
|  | Computing Interface:           | 1x Intel NUC i5, 256GB, 8GB RAM                                                                                     |
|                                                                                       | MaxSpeed:                      | 7" touchscreen, power button, break-release, USB port, LAN port                                                     |
|                                                                                       |                                | 1.1 m/s                                                                                                             |
|                                                                                       |                                |                                                                                                                     |
| <b>TORSO</b>                                                                          |                                |                                                                                                                     |
|  | Dimensions (L/W/H):            | 43/80 cm                                                                                                            |
|                                                                                       | Weight:                        | 43,2 kg                                                                                                             |
|                                                                                       | Power Supply:                  | Power supply from base 48V<br>Internal levels 48V, 24V, 19V                                                         |
|                                                                                       | Communication:                 | Gigabit-Ethernet                                                                                                    |
|                                                                                       | Computing:                     | 1-3x Intel NUC i5, 256GB, 8GB RAM                                                                                   |
|                                                                                       | Interface:                     | 7" touchscreen                                                                                                      |
|                                                                                       | Lighting:                      | 58x independent RGB-LEDs in ring                                                                                    |
|                                                                                       |                                |                                                                                                                     |
| Torso joint 1-DOF:                                                                    |                                | Max Joint Vel: 200°/sec<br>Joint Limits: -360° to 360°                                                              |
| Torso joint 3-DOF:                                                                    |                                | Max Joint Vel: 200°/sec per axis --> ~ ca. 1 bow/sec<br>Joint Limits: -360° to 360° all axis                        |

# Care-o-Bot

- Household robot that assists people in performing their everyday life at home, such as cleaning table and clearing dish washer.
- Not limited to assisting older people



## TECHNICAL DATA SHEET



Care-O-bot 4

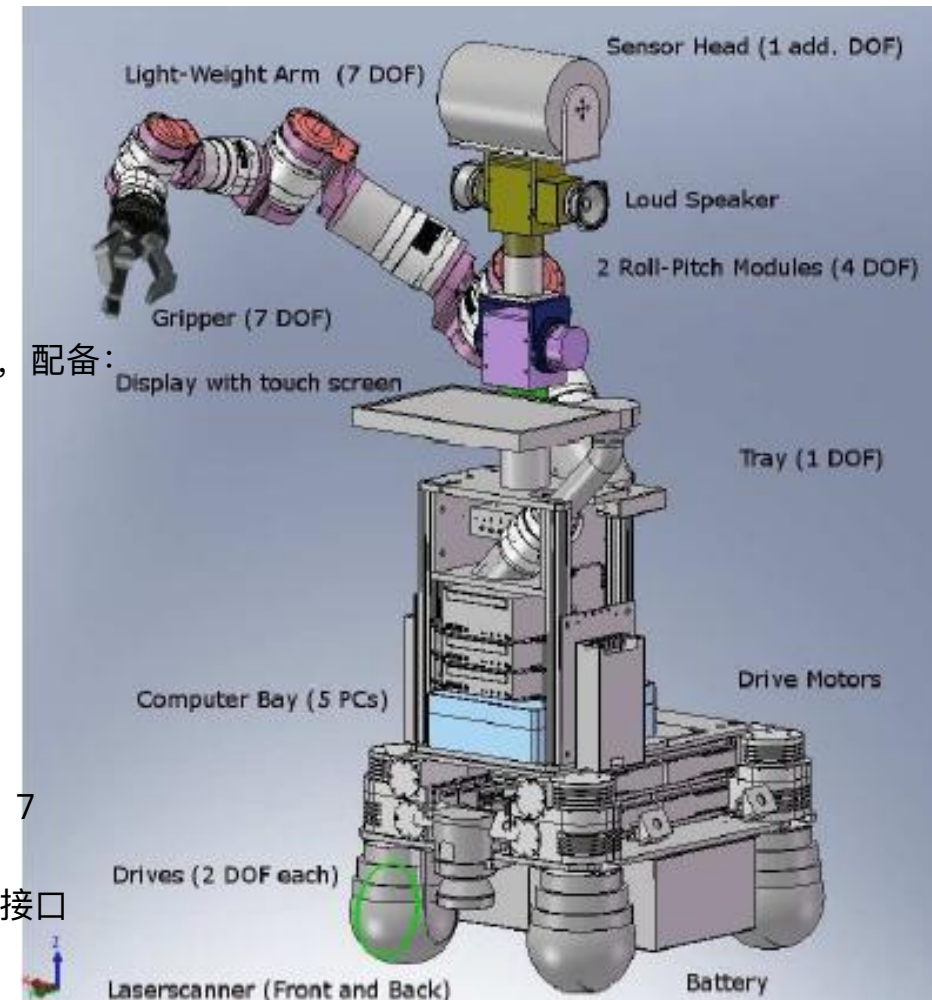


| MODULE                                                                                                 | COMPONENT/<br>PROPERTY                                                                                                                        | VALUE                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>ARM AND GRIPPER | <b>ARM</b><br>Reach:<br>Weight:<br>Degrees of freedom:<br>Communication:<br>Motors:<br>Payload:                                               | ca. 90 cm (from center of first ball)<br>13,9 kg<br>7<br>CANOpen<br>Schunk PowerBalls (PRL100, ERB145, ERB115)<br>5kg                                                                                                                                                             |
|                                                                                                        | <b>GRIPPER</b><br>Dimensions (L/H/W):<br>Weight:<br>Degrees of freedom:<br>Communication:<br>Camera:<br>Light:<br>Laser:<br>Other Interfaces: | 17/12,5/11 cm<br>0,9 kg<br>2<br>USB<br>3D-RGBD camera (range 15-200 cm)<br>1x 3W Daylight LED, 1x 1W (per channel) RGB-LED<br>Laserpointer, red, <5mW<br>6x configurable GPIO-ports<br>1x 20MHz SoC with logic blocks, USB, free programmable<br>1x USB port for external sensors |
| <br>HEAD            | Dimensions (D/H):<br>Weight:<br>Communication:<br>Computing:<br>Interface:                                                                    | 37/51 cm<br>14,7 kg<br>Gigabit Ethernet<br>1x Intel NUC i5, 256GB, 8GB RAM<br>15" touchscreen, 2x 20W speaker, 1x mic                                                                                                                                                             |
|                                                                                                        | Head joint 1-DOF:<br>Head joint 3-DOF:                                                                                                        | Max Joint Vel: 200°/sec<br>Joint Limits: -360° to 360°<br>Max Joint Vel: 200°/sec per axis<br>Joint Limits: -360° to 360° all axis                                                                                                                                                |
| <br>SENSOR RING   | Dimensions (DH):<br>Weight:<br>Communication:<br>Computing:<br>Speed:<br>Angle Limits:                                                        | 35/9cm<br>1kg + custom sensors<br>Gigabit Ethernet<br>1x Intel NUC i5, 256GB, 8GB RAM<br>200°/sec<br>-360° to 360°                                                                                                                                                                |

# 硬件结构

## Care-o-Bot hardware architecture

- 28 DOF
  - 28个自由度
  - 4个全向轮，方向和速度独立控制，实现全向移动
- 4 wheels, indepently controlled in orientation and speed, so that the mobile base is omnirdirectional
- 600x600x340mm mobile base, with:
  - Battery pack Li-Io (50 V, 60 Ah)
  - 2 laser scanners
  - Navigation PC
- The torso supports sensors, arm and tray. It contains most electronic components and control PCs
  - 移动底盘尺寸 600x600x340 mm, 配备:
    - 锂离子电池组 (50 V, 60 Ah)
    - 2台激光扫描仪
    - 导航PC
- Total height: 770 mm
- Robotic arm: Schunk LWA3, 7-DOF
  - 总高 770 mm
  - 机械臂: Schunk LWA3, 7 自由度
  - 支持多种末端执行器快换接口
- Attachment for diverse end effectors



# 模块化设计

## Care-o-Bot modular design

- 可配置1条、2条或完全不装机械臂
- 颈部和腕部的专利球形关节为选配
- 若用于送饮品，一只手可替换为托盘，或移动底盘单独作为送餐推车使用
- The robot can be equipped with one, two or no arms at all.
- The patented spherical joints on the robot's neck and hips are optional.
- If the intended purpose is to serve drinks, one hand can be replaced by a tray, or the mobile base platform can be used on its own as a serving trolley.
- Individual robot platforms can be configured for a wide range of applications: a mobile information center in museums, DIY stores and airports, for collection and delivery services in homes and offices, for security applications or as museum robots at attractions
- 单个机器人平台可配置为多种用途：博物馆/家装超市/机场的移动信息中心、家庭与办公室的物料配送、安防应用、景点导览机器人



As a roboticist, how difficult do you see the following tasks:

✓ 0

Robot navigation in an apartment

30%

Robot recognizing common objects

40%

Robot grasping common objects

60%

Robot understanding user's requests

70%

Robot handing over objects to the user

80%

# Summary

- Brief recall of target users
- Classification of manipulation aids
  - High-tech vs. low-tech
  - Specialized vs. general use
- Examples of specialized AMRs (e.g., feeders)
- ARMs – generalities on control
  - Fixed workstations
  - Wheelchair-mounted manipulators
  - Mobile robots

## Questions ?

- 操作辅助器具的目标用户回顾
- 操作辅助器具的分类
  - 高技术 vs 低技术
  - 专用型 vs 通用型
- 专用型操作辅助器具实例（如喂食机器人）
- 辅助机械臂（ARMs）——控制方式概述
  - 固定式工作站
  - 轮椅搭载式机械臂
  - 移动机器人